Introduction and concepts in community health

Goal: Give overview of community health

Specific objectives:

- Define the key concepts and terms used in community health
- Explain determinants of health
- Describe community entry process

- **Community** - a group of people who share some important feature of their lives and use some common agencies and institutions. Community is a specific group of people, often living in a defined geographical area, who share a common culture, values and norms and are arranged in a social structure according to relationships that the community has developed over a period of time.

- **Health** - a balanced state of well-being resulting from harmonious interactions of body, mind, and spirit."

- **Community health -** is defined by meeting the needs of a community by identifying problems and managing interactions within the community

- **Population** – a group of people living in a geographical setting in a period of time. Population can be described in terms of:
    - Cohort – group of people born in same period. They grow together in same age group
    - Risk group - Some population groups are at considerably higher risk of contracting disease than others. These include infants, children under five years of age, pregnant women, patients with HIV/AIDS, as well as non-immune migrants, mobile populations and travellers.
    - Density – is the number of people living in a square kilometer

- **Needs for health services** – Health services such as immunization, examination, treatment etc. are basic needs of every individual, family, community or entire population.

- **Demand for health services -** Every individual, family, community or entire population has a right to demand for essential and quality health services.

- **Primary health care** – Are interventions that focus on the individual or family. Are essential health services such as immunization, health education, nutrition and food supply, treatment and control of minor illnesses etc.

- **Risk of diseases –** Exposure to a disease. Some risk exposures are drug and alcohol consumption, immune-suppression etc.

Aims - improving the health of a community range from improving access to medical care to public health communications campaigns. The aims of community health are:

- To bring essential health services closer to the community members
- Empower community members to initiate, implement and own activities promoting their health

Determinants of health

- Economic determinants including income, employment and working conditions;
- Social determinants including social support, safety in the home and community, participation in civic activities and healthy child development; education and literacy

- Physical environmental determinants including the state of the natural environment, the presence of environmental tobacco smoke, availability of transportation and affordable and adequate housing; and
- Other determinants such as personal health practices, health services and biology and genetic endowment.

Determinants of community health

- *Community development* - The community development process is based on the philosophical belief that people and communities are entitled to have control over factors that affect their lives

- *Group* **-** People who interact and share a common purpose or purposes. There is no clear distinction between a group and a community except that groups tend to have fewer members than a community

- *Community health standards* - Promoting health; Building individual and community capacity through engagement and participation; Building relationships and ownership; Facilitating access and equity; Demonstrating professional responsibility, accountability and sustainability.

- *Community Engagement* **-** is the process of working collaboratively with and through groups of people affiliated by geographic proximity, special interest, or similar situations to address issues affecting the well-being of those people. It is a powerful vehicle for bringing about environmental and behavioral changes that will improve the health of the community and its members. It often involves partnerships and coalitions that help mobilize resources and influence systems, change relationships among partners, and serve as catalysts for changing policies, programs, and practices

- *Cultural Influence* **-** Health behaviors are influenced directly by elements of one's culture. As a result, social norms and other elements of community culture provide a potential tool for disease prevention and health promotion. Culture involves "the integrated pattern of human knowledge, belief, behavior, and material traits characteristic of a social group". An individual's culture influences his or her attitude toward various health issues, including perceptions of what is and is not a health problem, methods of disease prevention, treatments for illness, and use of health providers.

- *Community Participation* **-** Concepts concerning community participation offer one set of explanations as to why the process of community engagement might be useful in addressing the physical, interpersonal, and cultural aspects of individuals' environments.

- *Community Empowerment* **-** Mobilizing and organizing individuals, grass-roots and community-based organizations, and institutions, and enabling them to take action, influence, and make decisions on critical issues. It is important to note, however, that no external entity should assume that it can bestow on a community the power to act in its own self-interest. Empowerment takes place at three levels:
    - Individual,
    - Organizational or group, and
    - Community levels.

- *Capacity Building* **-** Another set of organizing concepts that can help guide approaches to effective engagement involves the process of capacity building. Individuals and organizations can gain control and influence and become players and partners in community health decision-making and

action, they may need resources, knowledge, and skills above and beyond those they already bring to a particular problem

- *Community Organization* **-** A "process by which community groups are helped to identify common problems or goals, mobilize resources, and in other ways develop and implement strategies for reaching goals they have set

Community health elements are:

- Promotion of healthful living
- Prevention of health problems
- Treatment of disorders
- Rehabilitation
- Evaluation and
- Research

Community health services are:

- Child health services (Immunization and Growth monitoring)
- Maternal services (Family planning, Antenatal, Postnatal, Health education)
- Treatment of minor illnesses
- Health promotion
- Referrals and follow up

Community health standards

- Engagement
- Participation
- Ownership
- Sustainability

Community entry

- Refers to the process of initiating, nurturing and sustaining a desirable relationship with the purpose of securing and sustaining the community's interest in al aspect of a programme.
- The process, principles and techniques of community mobilisation and participation. This involves recognizing the community its leadership and people and adopting the most appropriate process in meeting, interacting and working with them.
- Process involve:
  - Knock and enter upon response.
  - First greet chief and elders. Introduce yourself to the chief elders and other local leaders.
  - Inform them of your work with them.
  - Ask for their permission and advice and state your mission e.g. getting information about the village or introduction of new health programme.
  - Thank them for their co-operation.
  - Identify contact persons

Millennium Development Goals

The Millennium Development Goals (MDGs) are eight international development goals that were established following the Millennium Summit of the United Nations in 2000, following the adoption of the United Nations Millennium Declaration. All 189 United Nations member states at the time (there are 193 currently) and at least 23 international organizations committed to help achieve the following Millennium Development Goals by 2015:

1. To eradicate extreme poverty and hunger

2. To achieve universal primary education
3. To promote gender equality and empower women
4. To reduce child mortality
5. To improve maternal health
6. To combat HIV/AIDS, malaria, and other diseases
7. To ensure environmental sustainability
8. To develop a global partnership for development

Problem solving in community medicine

- What is a problem?
  - Problems are *opportunities* to make some good things happen. If it weren't for problems, what would be our motivation to create change?
  - Problems are also *challenges*. They call upon the best of our abilities, and ask us to go beyond what we thought we could do. They make life interesting, and, at least sometimes, fun. Without problems, life could be pretty boring.

- **Problem solving process:**

  Step 1 - Problem Definition
  Before you are ready to take any steps to solve the problem, you first have to be sure that you are clear about what the problem really is. It can be easy to get distracted by solving a different problem than what is actually causing distress if it is easier than dealing with the real problem.
  This step involves thinking about the following questions:
  - How is the current situation different from what I actually want it to be?
  - What do I actually want, or how do I actually want things to be?
  - What is preventing me from achieving my goals, or from things being the way I want them to be?

  Step 2 – Problem Analysis
  Once you have defined the problem, you need to think about it from different perspectives to insure that you understand all the dimensions of the problem.
  The following questions can be useful to help you analyze the problem.
  1. How is this problem affecting me?
  2. How is this problem affecting other people?
  3. Who else is experiencing this problem?
  4. How do other people deal with this problem?

  STEP 3 – Establish your goals
  Once you have looked at the problem from different perspectives, you can decide what you want to achieve and establish your goals. You need to answer the very specific question – "What is my immediate goal?"
  Examples:
  - Improve my time management skills
  - Complete assignments on time
  - Improve my grades

  STEP 4 – Generate possible solutions
  - During this stage the goal is to generate as many possible solutions as you can.

- Do not worry about whether or not they are realistic, practical, or effective.
- Be sure to write down all the possibilities you generate so that you can approach this task systematically.

STEP 5 – Analyze the solutions

During this stage, you will examine each alternative and write down both the advantages and disadvantages to each. Some considerations to keep in mind include:

- Is it relevant to my situation?
- Is it realistic?
- Is it manageable?
- What are the consequences – both good and bad?
- What is the likelihood that it is going to help me reach my goal?

STEP 6 - Implementation

The last step is to implement the solution you have chosen. This step involves identification of all the steps necessary to implement it, and also on-going monitoring of the effectiveness of the solution to make sure that it actually solved the problem. During this stage of the process, ask yourself the following questions:

1. How effective is the solution?
2. Did it achieve what I wanted?
3. What consequences (good and bad) did it have in my situation?

Factors Contributing to the Success of Community Engagement Efforts

- Environmental
- History of collaboration or cooperation in the community
- Membership - Mutual respect, self-interest, understanding, and trust
- Process/Structure - Members feel ownership — share stake in both process and outcome
- Every level in each organization in collaborating groups participates in decision-making
- Flexibility of collaborating group
- Clarity of roles and guidelines
- Communication - Open and frequent interaction, information, and discussion; Informal and formal channels of communications
- Purpose - Goals clear and realistic to all partners; Shared vision;
- Resources - Sufficient funds; Skilled convener

Communicable Diseases

Goal: Understand the concepts of communicable and non-communicable disease

Specific objectives:

- Discuss communicable diseases in relation to their causation, prevention and control
- Discuss prevention and treatment of non-communicable disease

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What are communicable diseases?

- A communicable disease such as a cold is a disease that spreads from person to person. Its also known as infectious or contagious
- Communicable diseases are diseases that you can "catch" from someone or something else.
- Some people may use the words contagious or infectious when talking about communicable diseases.
- Communicable diseases spread from one person to another or from an animal to a person. The spread often happens via airborne viruses or bacteria, but also through blood or other bodily fluid.
- Examples of communicable disease are:
  - HIV/AIDS
  - Influenza
  - Malaria
  - Polio
  - Tuberculosis
  - Hepatitis

What do you "catch" when you get a communicable disease?

- When a person becomes sick with a communicable disease it means a germ has invaded their body.
- Germs fear soap and water.
- Washing your hands well and often is the best way to beat these tiny warriors.

What are germs?

- Germs are tiny organisms (living things) that may cause disease. Germs are so small and sneaky that they creep into our body without being noticed. In fact, germs are so tiny that you need to use a microscope to see them. We don't know what hit us until we have symptoms (runny nose, cough, sore throat, fever, etc.)
- Germs are microbes. A microbe is a tiny living organism that can only be seen with a microscope. Microbes are the smallest form of life on Earth. Although some microbes can make you sick or may even kill you, most are harmless, and some are extremely helpful. Microbes can be found virtually anywhere - in air, water, plants, animals and humans. A Germ is a microorganism that causes disease. Germs are also known as pathogens.
- Broad example of germs are:
  - Bacteria – are everywhere. They live in sub-freezing cold temperatures and boiling hot temperatures. They live up high in the sky and down deep below the ocean. In fact scientists have found colonies of bacteria thriving 1,600 feet below sea level without oxygen or sunlight. Scientists also think there are signs of microbial life on Mars! Some bacteria are harmful but most are harmless or even helpful.

  - Viruses – depend upon a host to survive, grow, and reproduce. A host is the plant or animal on which or in in which another organism lives. Viruses can't live outside of other living cells. Once

they are inside the body, viruses can spread and make people sick. They reproduce by injecting their genes into a cell to produce thousands of new viruses. Viruses cause chicken pox, measles, flu, and many other diseases.

- Fungi - are multi-celled plant-like organisms. Unlike other plants, fungi aren't able to produce their own food from soil, water, and air. Instead, fungi get their nutrition from plants, food, and animals. Fungi thrive in warm, damp environments. Most fungi are not dangerous. An example of something caused by fungi is athlete's foot - that itchy rash that teens and adults sometimes get between their toes.

- Protozoa - are one-celled organisms like bacteria. Protozoa also love moisture and often spread diseases through contaminated water. Some protozoa cause intestinal infections.

Disease outbreak

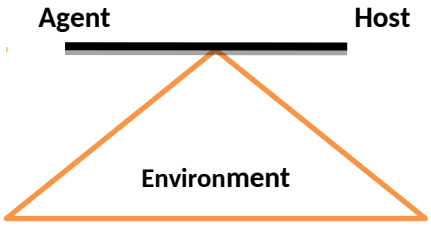
- A disease outbreak is the occurrence of cases of disease in excess of what would normally be expected in a defined community, geographical area or season. An outbreak may occur in a restricted geographical area, or may extend over several countries. It may last for a few days or weeks, or for several years. All disease outbreaks must be investigated.

REMEMBER - ALL microbes are NOT Germs!

- We could not digest our food without microbes.
- Garbage would not decay without microbes.
- Plants would not grow without microbes.
- The gas we pass is caused by bacteria in our intestines.
- Without microbes there would be no life on earth.

Epidemiologic triangle model of disease causation

- The epidemiologic triangle or triad is the traditional model of infectious disease causation
- It has 3 components:
  - External agent
  - Susceptible host
  - Environment that brings the host and agent together
- Environment influences the agent, the host and the route of transmission of the agent from a source to the host



Agent factors

- Originally refers to an infectious microorganism – virus, bacterium, parasite, or other microbe
- These agents must be present for a disease to occur. That is, they are necessary but not sufficient to cause a disease

Host factors

- Are intrinsic factors that influence individual's exposure, susceptibility, or response to a causative agent

- Age, race, sex, socio-economic status and behaviors (smoking, drug abuse, lifestyle, sexual practices and contraception, eating habits) are some of the many host factors which affect a person's likelihood of exposure

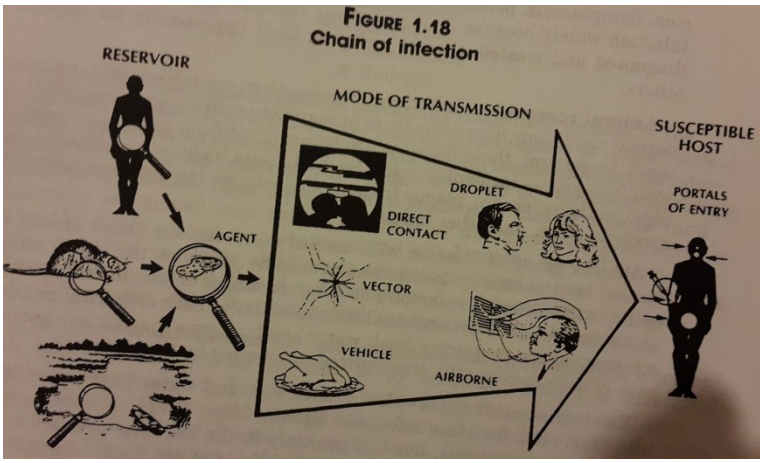
Environmental factors

- Are extrinsic factors which affect the agent and the opportunity for exposure
- It includes physical factors such as geology, climate, physical surrounding, biological factors such as insects that transmit agents, socio-economic factors such as crowding, sanitation etc.
- Agents, host and environmental factors interrelate in a variety of complex ways to produce a disease in humans.

Transmission cycle

Chain of infection

- Transmission occurs when the agent leaves its reservoir or host through a portal of exit, and is conveyed by some mode of transmission, and enters through an appropriate portal of entry to infect a susceptible host.



Reservoir

- Reservoir of an agent is the habitat in which an infectious agent normally lives, grows and multiplies
- This includes humans, animals and the environment
- The reservoir may or may not be the source from which an agent is transferred to a host

Human reservoir

- Many of the common infectious diseases have human reservoirs.
- Disease that are transmitted from person to person without intermediaries include sexually transmitted diseases, measles, mumps, streptococcal infections, most respiratory pathogens etc.
- Smallpox was eradicated after the last human case was identified and isolated because humans were the only reservoir for the smallpox virus
- Two types of human reservoir are:
    - Persons with symptomatic illness - Symptomatic persons are usually less likely to transmit infection widely because their symptoms increases their likelihood of being diagnosed and treated
    - Carriers - A carrier is a person without apparent disease who is nonetheless capable of transmitting the agents to others. Carriers may be asymptomatic carriers, who never show symptoms during the time they are infected, or may be incubatory or convalescent carriers who are capable of transmission before or after they are clinically ill

Animal reservoir

- Infectious diseases that are transmittable under normal conditions from animal to humans are called zoonoses. Such disease include brucellosis (cows and pigs), anthrax (sheep), plague (rodents), rabies (bats, dogs)

Environmental reservoir

- Plants, soil and water are also reservoir for some infectious disease. Many fungal agents such as those causing histoplasmosis live and multiply in the soil

Portal of exit and entry

- Is the path by which an agent leaves the source host or enters
- Tubercle bacilli and influenza viruses exit the respiratory tract; schistosome through urine; cholera vibrios through feces etc

Modes of transmission

- After an agent exists its natural reservoir, it may be transmitted to a susceptible host in numerous ways. These modes of transmission are classified as:
    - Direct (Direct contact and direct spread) - There is immediate transfer of the agent from a reservoir to a susceptible host by direct contact or droplet spread. It occur through kissing, skin-to-skin contact. Also refer to contact with soil or vegetation harboring infectious organisms

    - Indirect (Airborne, Vehicle borne, Vector borne) – This can be mechanical or biologic. An agent is carried from a reservoir (a susceptible host) by suspended air particles or by animate (vector) or inanimate (vehicle) intermediaries. Most vectors are arthropods such as mosquitoes, fleas and ticks. This is in contrast of instances in which an agent undergoes part of its life cycle inside a vector before being transmitted to a new host. When the agent undergoes changes within the vector, the vector is serving as both intermediate host and a mode of transmission. This type of direct transmission is called a biologic transmission

- Vehicles that may directly transmit an agent include food, water, blood, handkerchiefs, bedding etc

Principles of managing communicable diseases (Investigation and control of epidemics)

- Health departments investigate suspected outbreaks for various reasons:
    - Need to institute control and prevention measures
    - Opportunity for research and training
    - Program considerations
    - Public relations
    - Political concern
    - Legal obligations
- Principles are:
    - Outbreak occurs in any community
    - Outbreaks commonly occur in poor communities
    - It adds to burden of poverty
    - Prevention occurrences of outbreak is important
    - Spread of outbreak should be limited
    - Reduce harm of outbreaks to individual
    - Management of outbreak in competent way
    - Stop the victim from injury of the outbreak
    - Break the chain of infection

- Steps of outbreak investigation:
    - Prepare for fieldwork – Prepare before leaving for the field. Preparation can be grouped into 3 categories:

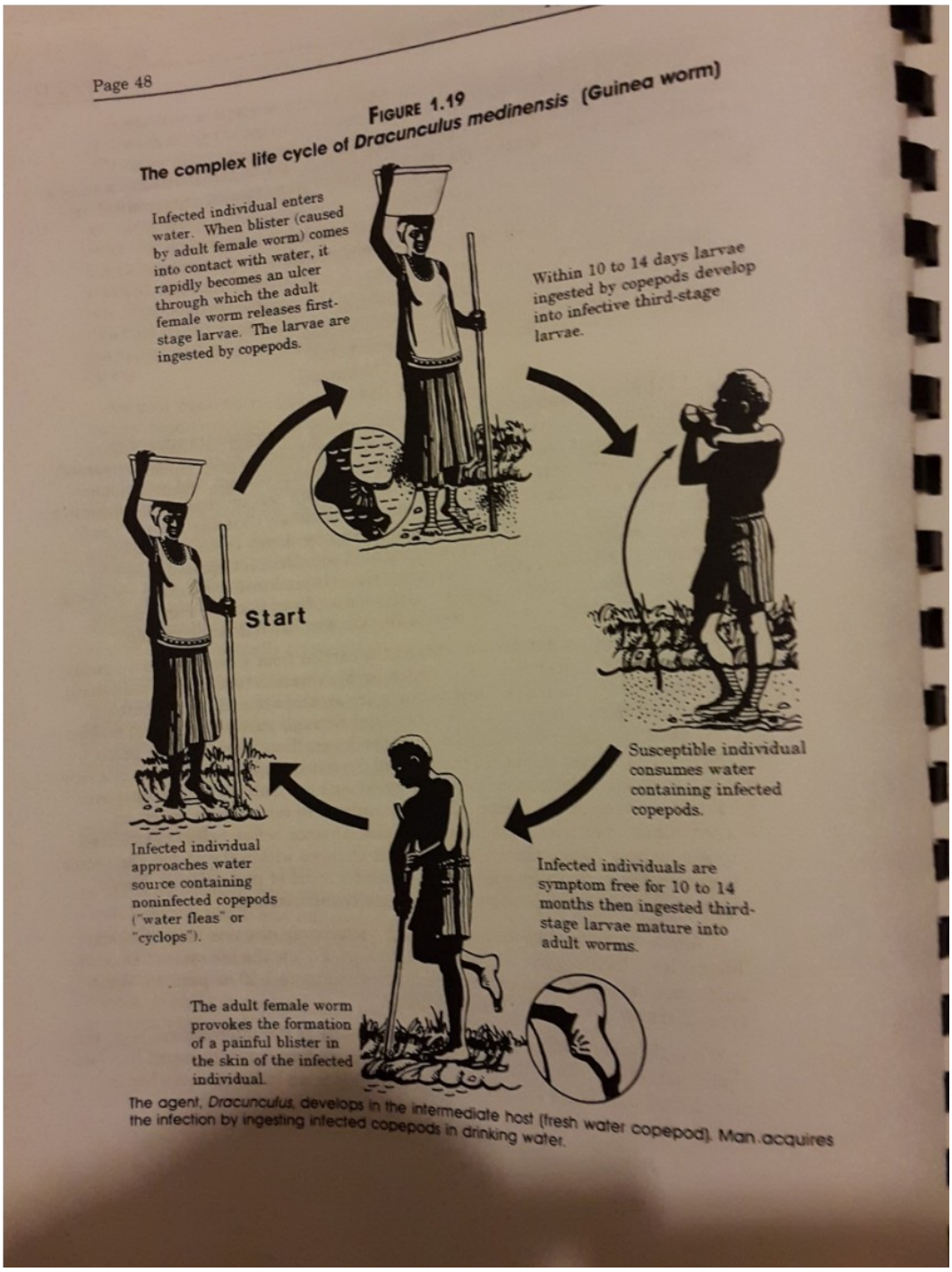
        - Investigation – have scientific knowledge, supplies and equipment for investigation, review literature
        - Administration – pay attention to administrative procedures e.g. get approvals
        - Consultation – know your role in the field. Consult with local staff/people
    - Establish the existence of an outbreak
        - an outbreak or epidemic is the occurrence of more cases of disease than expected in a given area over a particular period of time
        - you can establish outbreak through regular analysis of data or report from health care providers
        - compare observed with expected cases
    - Verify the diagnosis
        - Establish what disease is occurring by proper diagnosis
        - Review clinical findings and laboratory results
        - Visit and talk to some patients to get better understanding of the clinical features
    - Define and identify cases
        - Establish a case definition
            - A case definition is a standard set of criteria for deciding whether an individual should be classified as having the health condition of interest
            - Classify case definitions as: definite case; probable case; possible case
        - Identify and count cases
            - Cases that prompt concern are often only a small and no representative fraction of the total number of cases
            - Ask case patients if they know anyone else with the same illness
            - Collect the following information about every case:
                - Identifying information
                - Demographic information
                - Clinical information
                - Risk factor information
                - Reporter information
    - Perform descriptive epidemiology - Characterizing an outbreak by variables such as time, place and person is called descriptive epidemiology. Develop hypothesis. Evaluate hypothesis by:
        - Use analytical epidemiology to test your hypothesis
        - You can also use case control
        - As necessary, reconsider/refine hypothesis and execute additional studies
    - Implement control and prevention measures
    - Document and communicate findings through oral briefs or written report

Methods of control and eradication of diseases

- There is an important distinction among the 3 terms, prevention, control, and eradication.
- *Control:* The reduction of disease incidence, prevalence, morbidity or mortality to a locally acceptable level as a result of deliberate efforts; continued intervention measures are required to maintain the reduction. Example: diarrheal diseases.
- *Elimination of infections:* Reduction to zero of the incidence of infection caused by a specific agent in a defined geographical area as a result of deliberate efforts; continued measures to prevent re-establishment of transmission are required. Example: measles, poliomyelitis.
- *Eradication:* Permanent reduction to zero of the worldwide incidence of infection caused by a specific agent as a result of deliberate efforts; intervention measures are no longer needed. Example: smallpox.

- *Extinction:* The specific infectious agent no longer exists in nature or in the laboratory. Example: none.

Figure2



Non communicable disease

- These are medical conditions of noninfectious or non-transmittable to another person. Currently, the NCD are the leading causes of death and disease burden worldwide
- The 4 main types of non-communicable diseases are cardiovascular diseases (like heart attacks and stroke), cancers, chronic respiratory diseases (such as chronic obstructed pulmonary disease and asthma) and diabetes.

Risk factors

All age groups and all regions are affected by NCDs. NCDs are often associated with older age groups, but evidence shows that 16 million of all deaths attributed to non-communicable diseases occur before the age of 70. Of these "premature" deaths, 82% occurred in low- and middle-income countries. Children, adults and the elderly are all vulnerable to the risk factors that contribute to non-communicable diseases, whether from unhealthy diets, physical inactivity, exposure to tobacco smoke or the effects of the harmful use of alcohol.

These diseases are driven by forces that include ageing, rapid unplanned urbanization, and the globalization of unhealthy lifestyles. For example, globalization of unhealthy lifestyles like unhealthy diets may show up in individuals as raised blood pressure, increased blood glucose, elevated blood lipids, and obesity. These are called 'intermediate risk factors' which can lead to cardiovascular disease, a NCD.

A **risk factor** is any attribute, characteristic or exposure of an individual that increases the likelihood of developing a disease or injury. The following are risk factors:

- *Modifiable behavioral risk factors* - Tobacco use, physical inactivity, unhealthy diet and the harmful use of alcohol increase the risk of NCDs.

- *Metabolic/physiological risk factors* - These behaviors lead to four key metabolic/physiological changes that increase the risk of NCDs: Raised blood pressure, Overweight/obesity, Hyperglycemia (high blood glucose levels) and Hyperlipidemia (high levels of fat in the blood).

Risks classification

High risk includes any of the following:

- Percutaneous (e.g., needle stick) or mucous membrane exposure to blood or body fluids of a person with Ebola while the person was symptomatic
- Exposure to the blood or body fluids (including but not limited to feces, saliva, sweat, urine, vomit, and semen) of a person with Ebola while the person was symptomatic without appropriate personal protective equipment (PPE)
- Processing blood or body fluids of a person with Ebola while the person was symptomatic without appropriate PPE or standard biosafety precautions
- Direct contact with a dead body without appropriate PPE in a country with widespread transmission or cases in urban areas with uncertain control measures.
- Having lived in the immediate household and provided direct care to a person with Ebola while the person was symptomatic

Some risk includes any of the following:

- In countries with widespread transmission or cases in urban areas with uncertain control measures:
    - direct contact while using appropriate PPE with a person with Ebola while the person was symptomatic or with the person's body fluids
    - any direct patient care in other healthcare settings
- Close contact in households, healthcare facilities, or community settings with a person with Ebola while the person was symptomatic
    - Close contact is defined as being for a prolonged period of time while not wearing appropriate PPE within approximately 3 feet (1 meter) of a person with Ebola while the person was symptomatic

Low (but not zero) risk includes any of the following:

- Having been in a country with widespread transmission or cases in urban areas with uncertain control measures within the past 21 days and having had no known exposures
- Having brief direct contact (e.g., shaking hands) while not wearing appropriate PPE, with a person with Ebola while the person was in the early stage of disease
- Brief proximity, such as being in the same room for a brief period of time, with a person with Ebola while the person was symptomatic
- In countries without widespread transmission or cases in urban settings with uncertain control measures: direct contact while using appropriate PPE with a person with Ebola while the person was symptomatic or with the person's body fluids
- Traveled on an aircraft with a person with Ebola while the person was symptomatic

No identifiable risk includes:

- Contact with an asymptomatic person who had contact with person with Ebola
- Contact with a person with Ebola before the person developed symptoms
- Having been more than 21 days previously in a country with widespread transmission or cases in urban areas with uncertain control measures
- Having been in a country with Ebola cases, but without widespread transmission or cases in urban settings with uncertain control measures, and not having any other exposures as defined above
- Having remained on or in the immediate vicinity of an aircraft or ship during the entire time that the conveyance was present in a country with widespread transmission or cases in urban areas with uncertain control measures, and having had no direct contact with anyone from the community.

Prevention and control strategies

- A comprehensive approach is needed that requires all sectors, including health, finance, foreign affairs, education, agriculture, planning and others, to work together to reduce the risks associated with NCDs, as well as promote the interventions to prevent and control them.
- To focus on lessening the risk factors associated with these diseases. Low-cost solutions exist to reduce the common modifiable risk factors (mainly tobacco use, unhealthy diet and physical inactivity, and the harmful use of alcohol) and map the epidemic of NCDs and their risk factors.
- High impact essential NCD interventions that can be delivered through a primary health-care approach to strengthen early detection and timely treatment. These measures can be implemented in various resource levels.
- Lower-income countries generally have lower capacity for the prevention and control of non-communicable diseases. Countries with inadequate health insurance coverage are unlikely to provide universal access to essential NCD interventions.
- Prevention is NCD is in three levels:
    - Primary prevention - is aimed at maintaining a healthy population; i.e., preventing the occurrence of disease. It involves the healthy population.
    - Secondary prevention - (also called disease control) attempts to minimize resultant damage after disease has occurred and primary prevention has failed.
    - Tertiary prevention - consists of rehabilitation after primary and secondary prevention has failed.
- Secondary and tertiary Prevention is aimed at the diseased or patient population.
- Preventive approaches are:
    - Naturalizing the reservoir
    - Reducing contact potential
    - Increasing resistance

Socioeconomic impacts of NCDs

- NCDs threaten progress towards the UN Millennium Development Goals and post-2015 development agenda.
- Poverty is closely linked with NCDs. The rapid rise in NCDs is predicted to impede poverty reduction initiatives in low-income countries, particularly by increasing household costs associated with health care.
- Vulnerable and socially disadvantaged people get sicker and die sooner than people of higher social positions, especially because they are at greater risk of being exposed to harmful products, such as tobacco or unhealthy food, and have limited access to health services.
- In low-resource settings, health-care costs for cardiovascular diseases, cancers, diabetes or chronic lung diseases can quickly drain household resources, driving families into poverty. The exorbitant costs of NCDs, including often lengthy and expensive treatment and loss of breadwinners, are forcing millions of people into poverty annually, stifling development.
- In many countries, harmful drinking and unhealthy diet and lifestyles occur both in higher and lower income groups. However, high-income groups can access services and products that protect them from the greatest risks while lower-income groups can often not afford such products and services.

Basic treatment

Common **-** There are numerous common communicable diseases, such as throat colds and the flu. They often produce symptoms like sore throat, runny nose, coughing, sneezing, headache, fever and nausea. These can typically be treated with adequate rest and over-the-counter medication. Medication for common communicable diseases may include pain relievers like Tylenol as well as nasal decongestants, cough syrups, anti-nausea pills and throat drops. Common communicable diseases usually dissipate over the course of a few days or weeks.

Chronic - Chronic communicable diseases are severe and not usually spread from person to person via airborne pathogens. They may be spread via parasites, sexual contact or transmission of bodily fluids like blood. Unlike throat colds or other common illnesses, chronic communicable diseases may last months or even a lifetime. Examples of chronic communicable disease include malaria, Lyme disease, diphtheria and hepatitis. Treatment for chronic communicable diseases usually involves some form of hospitalization, intravenous antibiotics, prescription medication and intensive care.

Vaccines - One effective way to treat a communicable disease is by receiving a vaccine before your body becomes infected. Vaccines work by infecting your body with a weakened strain of the actual disease. Your body then produces antibodies to fight the disease. Antibodies remain in the system after the weakened strain is gone and protect the body from later infections. Certain vaccines need only to be administered once, such as vaccines for mumps, measles, whooping cough and polio. Others, like the flu vaccine, should be taken annually.

Prevention/Solution - Prevention is the most effective form of treatment for communicable disease. Boost your immune system by incorporating high levels of antioxidants, fruits, vegetables and antihistamines into your diet. Engage in at least 30 minutes of exercise daily to maintain strength and proper circulation. Drink high amounts of water (in excess of two liters per day) and maintain adequate rest for your body. Wash your hands with soap and water after using the bathroom, handling uncooked meat, sneezing, coughing or touching a dirty surface.

Warning - Some communicable diseases have no cure. For example, there is no cure for sexually transmitted diseases (STDs) like herpes and HIV. Avoiding sexual contact with individuals infected with STDs is the best and only way to ensure that you do not become infected with such communicable diseases

Community participation in disease control program

The determinants of community participation in disease control are:

- Political background,
- Community characteristics,
- Managerial capacity of the provider
- Epidemiology of the disease.

Recently, community participation in organized efforts to control Chagas' disease has become more dynamic, with increasingly active involvement by local community members.

Community participation per se has inherent value because of its positive effect on social relationships and community solidarity.

***** Outline evidence based community participation in the control of communicable disease*****

Disease Prevention, control and eradication

Goal: Improve knowledge and skills on disease prevention

Specific objectives:

- Explain the purposes and levels of disease prevention
- Explain the approaches and challenges to disease prevention

Prevention – is defined as intervention or interventions that interrupt the web of causality leading to one or more aspects of ill health.

Preventive healthcare – these are measures taken for disease prevention, as opposed to disease treatment. It is also called preventive medicine or prophylaxis. Each year, millions of people die preventable deaths.

Purpose of disease prevention

- Prevention is better than cure
- Stop new cases of disease from occurring
- Stop existing cases of diseases from spreading

Primary level prevention

- Here the goal is to protect healthy people from developing a disease or experiencing an injury in the first place. For example:
  - education about good nutrition, the importance of regular exercise, and the dangers of tobacco, alcohol and other drugs
  - education and legislation about proper seatbelt and helmet use
  - regular exams and screening tests to monitor risk factors for illness
  - immunization against infectious disease
  - controlling potential hazards at home and in the workplace

Secondary level prevention

- These interventions happen after an illness or serious risk factors have already been diagnosed.
- The goal is to halt or slow the progress of disease (if possible) in its earliest stages; in the case of injury, goals include limiting long-term disability and preventing re-injury. For example:
  - telling people to take daily, low-dose aspirin to prevent a first or second heart attack or stroke
  - recommending regular exams and screening tests in people with known risk factors for illness
  - providing suitably modified work for injured workers

Tertiary level prevention

- This focuses on helping people manage complicated, long-term health problems such as diabetes, heart disease, cancer and chronic musculoskeletal pain.
- The goals include preventing further physical deterioration and maximizing quality of life. For example:
  - cardiac or stroke rehabilitation programs
  - chronic pain management programs
  - patient support groups

Preventive techniques

- Prevention is the first line of defense against disease.
- At least 8 preventive techniques are available for disease prevention:
  - ✓ Exclusion - of causative agents of disease from specific geographic areas, or quarantine.
  - ✓ Control methods - such as immunization, environmental control, and chemical agents to protect specific animal populations from endemic diseases, diseases normally present in an area.
  - ✓ The mass education - of people about disease prevention.
  - ✓ Early diagnosis - of illness among members of an animal population is important so that disease manifestations do not become too severe and so that affected animals can be more easily managed and treated.
  - ✓ Recognition - of the relationship among high risk population
  - ✓ Health protection measures - and the second group is the health promotion measures. The two approaches are complementary and interactive, and it is unlikely that one would be effective without the other. A third group of measures include preventive health services.
  - ✓ Health promotion measures - include activities that individuals and communities can use to promote healthy life style. For example: public education programs, health warning labels, advertising and marketing, and media portrayals.

Simple things you can do to prevent illness.

- Keep Immunizations up to date
- Wash your hands often
- Be aware of what you eat, and be careful how you prepare it
- Use antibiotics exactly as prescribed
- Report to your doctor any worsening infection that does not get better after you take a prescribed antibiotic
- Be cautious around wild and domestic animals that are not familiar to you
- Avoid areas of insect infestation
- Avoid unsafe unprotected sex and injection drug use
- Stay alert to disease threats when you travel
- When sick, allow yourself time to heal and recover.

The strategic directions to disease prevention

- The strategic directions provide a strong foundation for all of our nation's prevention efforts and include core recommendations necessary to build a prevention-based strategy.
- Each Strategic Direction guides actions that will demonstrably improve health. Together, the Strategic Directions create the web needed to fully support in leading longer and healthier lives.
- The four Strategic Directions are:
  - Healthy and Safe Community Environments
  - Clinical and Community Preventive Services
  - Empowered People
  - Elimination of Health Disparities
- Within the Strategy, the priorities provide evidenced-based recommendations that are most likely to reduce the burden of the leading causes of preventable death and major illness.
- The Priorities are designed to improve health and wellness of the population, including those groups disproportionately affected by disease and injury.
- The seven Priorities are:
  - Tobacco Free Living
  - Preventing Drug Abuse and Excessive Alcohol Use

- Healthy Eating
- Active Living
- Injury and Violence Free Living
- Reproductive and Sexual Health
- Mental and Emotional Well-Being

A Plan for Better Health and Wellness

The overarching goal of the National Prevention Strategy is to increase the number of people who are healthy at every stage of life. The strategy provides evidence-based recommendations that are fundamental to improving the nation's health through the active engagement of all sectors of society to help achieve four broad strategic directions:

- Building Healthy and Safe Community Environments - Prevention of disease starts in our communities and at home; not just in the doctor's office. For example, businesses and employers can adopt practices to encourage their workforce to increase physical activity and reduce pollution (e.g., workplace flexibility, rideshare and vanpool programs, park-and-ride incentives, travel demand management initiatives, and telecommuting options).

- Expanding Quality Preventive Services in Both Clinical and Community Settings - When people receive preventive care, such as immunizations and cancer screenings, they have better health and lower health care costs. For example, expanding the linkages between clinical and community prevention efforts, such as diabetes prevention programs that support preventive efforts among underserved groups and can improve access to preventive services.

- Empowering People to Make Healthy Choices - Policies and programs can make healthy options the easy and affordable choice, and when people have access to actionable and easy-to-understand information and resources, they are empowered to make healthier choices. For example, health care professionals can use multiple communication tools (e.g., mobile phone applications, personal health records, and credible health websites) and culturally competent methods to support more traditional written and oral communication.

- Eliminating Health Disparities - By eliminating disparities in achieving and maintaining health, we can help improve quality of life for all Americans. For example, health care providers can train and hire more qualified staff from underrepresented racial and ethnic minority groups and people with disabilities.

Role of environment and individual approaches to disease prevention

- *Ecosystem* – Environment is where man, pathogens (virus, bacteria, fungi and protozoa) live. The interaction between man with environment enable transfer of germs to human system

- *Difficulties in changing behavior*- To prevent disease we increasingly ask people to begin do things that they have not done previously, to stop doing things that they have been doing for years and to do more of some things and less of others

- *Pattern of disease rate* - Some groups often have a characteristic over time even though individuals come and go from these groups. If groups have different rates over time, there may be something promotes or discourages disease among individuals in those groups

- *Socio-economic status*- A consistent finding dated from the twelfth century, is that people in the lowest socioeconomic groups have the highest rates of morbidity and mortality, whether the

socioeconomic status was studied in relation to education, income, or occupation: the lower the level the higher the death and morbidity rate.

- Marital status - It has been known for many years that people who are not married-whether single, separated, widowed or divorced- have higher mortality rates than married people.

- Gender - One of the most well-established facts among students of health and disease is that men have higher mortality rates than women. This excess of male deaths occurs at every age and for every major cause for which comparison is possible.

Health education

- Is any is any combination of learning experiences designed to help individuals and communities improve their health, by increasing their knowledge or influencing their attitudes.
- A level of education plays a major role in health education. Different levels of education has different levels of perception and understanding
- Target audiences are those that the education is intended for. They can be:
    - Primary target audience – people who are directly affected
    - Secondary target audience – people who are indirectly affected
    - Tertiary target audience – people who are indirectly affected and can influence policy
- Some of the education outlets in the community include chief's baraza, churches, schools, social places, boda-boda terminals etc.
- Promoting health and preventing disease in the population and in the wider community involves the following:
    - *Demonstrate effective, patient-centered counseling*
        - ✓ Interventions that address patients' personal health behaviors and practices are vitally important.
        - ✓ Clinicians can help individuals change their health-related behaviors by using efficient, patient-centered counseling strategies that are based on an understanding of how people change.

    - *Understand the principles of and procedures for screening*.
        - ✓ Another central health promotion/disease prevention strategy that students need to be skilled in is screening, or testing for health problems in the absence of symptoms or signs indicating the presence of such a problem.
        - ✓ Students need to learn (1) when screening is appropriate, (2) what procedures to use and how to use them effectively, and (3) how to interpret and use the results of screening tests.

    - *Describe the importance and use of immunization and prophylaxis*.
        - ✓ Understand the important role that widespread childhood immunization has played in drastically reducing the incidence of many illnesses, including diphtheria, pertussis, tetanus, poliomyelitis, measles, mumps, and rubella.
        - ✓ In addition, learn which childhood immunizations are essential, when and how to administer them, and potential risks.
        - ✓ Knowledge of the appropriate use of adult immunizations is also necessary.

- *Implement strategies to promote the health of the population.*
  - ✓ Although health care professionals traditionally have focused on individual patients who present themselves for care one at a time in the office or clinic, students preparing for practice in today's health care system also must learn to focus on the care of populations of patients.

- *Engage in public education and advocacy to prevent disease.*
  - ✓ For some health problems, preventive interventions aimed at the wider community may be especially appropriate and effective.
  - ✓ Such problems include unintended pregnancy in adolescents, youth violence, and domestic violence, initiation of tobacco use, sexually transmitted diseases, healthy dietary practices, and unintentional injuries.

The challenges to disease prevention

Political challenges

- The scope for the prevention of diseases is enormous.
- There will be scope for prevention of all types and at all levels, and much of this will require political change.
- Smoking-related diseases are increasing in developing countries where tobacco companies have been promoting sales.
- Widespread disease and death still occur in many countries which don't have a clean and plentiful water supply.
- Wars and armed conflict have hindered or destroyed the development of basic infrastructure in many parts of the world. Worldwide, it is evident that measures at government level will yield the greatest gains.

Specific challenges

- Many of these are clinical or individual challenges.
- This may include the barriers to prevention. Articles of particular note include:

Taxation and cost issues

- Consumption may fall so much that total revenue from that source falls. This can be politically challenging with industry concerned about economic damage.
- Most interventions have costs and do not have the advantages of revenue generation associated with, for example, taxation on tobacco. This includes screening programmes or advertising campaigns.
- Immunization programmes cost money to set up in private health institutions

Demographic issues

- A growing elderly population puts more strain on health and social services. A significant number of people now spend a quarter of their lives as pensioners retired from working life.

Personal barriers

- Cost and quality
- Information
- Exercise
- Social, educational and cultural barriers

Administrative barriers

Community Diagnosis

Goal: Improve skills and knowledge on community diagnosis

Specific objectives:

- Define the concepts of community diagnosis and its purposes
- Describe the steps in carrying out community diagnosis
- Conduct a participatory rural appraisal as a tool for community diagnosis

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Community – is a cluster of people with at least one common characteristic (geographic location, occupation, ethnicity, housing condition). It can also be defined as a group of people with a common characteristic or interest living together within a larger society

Diagnosis – is the identification and confirmation of a problem

Community sub systems - A community is a whole entity that functions because of the interdependence of its parts or subsystems. Community core are: history, socio-demographic characteristics, vital statistics, values/beliefs/religions. Community sub systems are: Physical environment; Education; Safety and transportation; Politics and government; Health and social services; Communication; Economics; and Recreation.

Community diagnosis

- Refers to the identification and quantification of health problems in a community as a whole in terms of mortality and morbidity rates and ratios, and identification of their correlates for the purpose of defining those at risk or those in need of health care.
- Is a comprehensive assessment of health status of the community in relation to its social, physical and biological environment
- According to WHO definition, it is "a quantitative and qualitative description of the health of citizens and the factors which influence their health. It identifies problems, proposes areas for improvement and stimulates action

Purpose - The purpose of community diagnosis is to define existing problems, determine available resources and set priorities for planning, implementing and evaluating health action, by and for the community. The following are the importance of community diagnosis:

- Analyze the health status of the community
- Evaluate the health resources, services, and systems of care within the community
- Assess attitudes toward community health services and issues
- Identify priorities, establish goals, and determine courses of action to improve the health status of the community
- Establish an epidemiologic baseline for measuring improvement over time.
- To act as a data reference
- To provide an overall picture of the local community and the residents' concerns
- To suggest priority areas for intervention and the feasible solutions
- To indicate the resource allocation and the direction of work plans
- It helps to find the common problems or diseases, which are troublesome to the people and are easily preventable in the community.
- It is a tool to disclose the hidden problems that are not visible to the community people but are being affected by them.
- It helps to access the group of underprivileged people who are unable to use the available facilities due to poverty, prevailing discriminations or other reasons.

- It helps to find the real problems of the community people which might not have perceived by them as problems.

Implementation framework and process

Types of information in community diagnosis and intervention sources

- The information collected in a community diagnosis is called health indicators
- Characteristics of indicators:
  - **Valid,** i.e., they should actually measure what they are supposed to measure;
  - **Reliable** and objective, i.e., the answers should be the same if measured by different people in similar circumstances;
  - **Sensitive,** i.e., they should be sensitive to changes in the situation concerned,
  - **Specific,** i.e., they should reflect changes only in the situation concerned,
  - **Feasible,** i.e., they should have the ability to obtain data needed, and;
  - **Relevant,** i.e., they should contribute to the understanding of the phenomenon of interest.

Classifications of indicators:

- Mortality indicators; Morbidity indicators ; Disability rates ; Nutritional status indicators; Health care delivery indicators ; Utilization rates ; Indicators of social and mental health ; Environmental indicators; Socio-economic indicators ; Health policy indicators ; Indicators of quality of life' Other indicators

Mortality indicators are: ○ Crude death rates ○ Specific death rates: age/disease ○ Expectation of life ○ Infant mortality rate ○ Maternal mortality rate ○ Proportionate mortality ratio ○ Case Fatality rate	Morbidity indicators are: ○ Incidence and prevalence ○ Notification rates ○ Attendance rates: out-patient clinics or health centers. ○ Admission and discharge rates ○ Hospital stay duration rates
Disability indicators are: ○ No. of days of restricted activity ○ Bed disability days ○ Work/School loss days within a specified period. ○ Expectation of life free of disability	Nutritional indicators are: ○ Anthropometrics measurements ○ Height of children at school entry ○ Prevalence of low birth weight ○ Clinical surveys: Anaemia, Hypothyroidism, Night blindness

Health care delivery indicators are: o Doctor / Population ratio o Doctor / Nurse ratio o Population / Bed ratio o Population / per health center	Utilization indicators are: o Proportion of infants who are fully immunized in the 1st year of life i.e..immunization coverage. o Proportion of pregnant women who receive ANC. o Hospital-Beds occupancy rate. o Hospital-Beds turn-over ratio
Social/mental health indicators are: o Suicide & Homicide rates o Road traffic accidents o Alcohol and drug abuse	Environmental health indicators are: o Measures of Pollution o The proportion of people having access to safe water and sanitation facilities o Vectors density
Socio-economic indicators are: o Rate of population increase o Per capita income o Level of unemployment o Literacy rates - females o Family size o Housing condition e.g. No. of persons per room	Health policy indicators are: o Proportion of resources spent on health services. o Proportion of funds spent on health related activities. o Proportion of total health resources devoted to primary health care

Community diagnosis process/steps

Implementation of community diagnosis takes stages:

- Initiation stage
  - Set up a dedicated committee or working group to manage and coordinate the project.
  - The committee should involve relevant parties such as government departments, health professionals and non-governmental organizations.
  - Identify the available budget and resources to determine the scope of the diagnosis.
  - Some of the common areas (scope) to be studied may include health status, lifestyles, living conditions, socioeconomic conditions, physical and social infrastructure, inequalities, as well as public health services and policies.
  - Once the scope is defined, a working schedule to conduct the community diagnosis, production and dissemination of report should be set.

- Data collection and analysis stage
  - Collect both quantitative and qualitative data.
  - Population census and statistical data (e.g. population size, sex and age structure, medical services, public health, social services, education, housing, public security and transportation, etc.) can provide background of the district.
  - Collect community data by conducting surveys through self-administered questionnaires, face to face interviews, focus groups and telephone interviews.
  - In order to ensure reliability of the findings, an experienced organization such as an academic institute can be employed for conducting the study.
  - The sampling method should be carefully designed and the sample size should be large enough to provide sufficient data to draw reliable conclusions.
  - Analyze and interpret the data.
  - Here are some practical tips on data analysis and presentation:
    - statistical information is best presented as rates or ratios for comparison
    - trends and projections are useful for monitoring changes over a time period for future planning
    - local district data can be compared with other districts or the whole population
    - graphical presentation is preferred for easy understanding

- Diagnosis stage

- Diagnosis of the community is reached from conclusions drawn from the data analysis.
- It should preferably comprise three areas:
  - Health status of the community
  - Determinants of health in the community
  - Potential for healthy community development

- Dissemination stage
  - The production of the community diagnosis report is not an end in itself
  - Efforts should be put into communication to ensure that targeted actions are taken.
  - The target audience for the community diagnosis includes policy-makers, health professionals and the general public in the community.
  - The report can be disseminated through the following channels:
    - ✓ Presentations at meetings of the health boards and committees, or forums organized, local community groups and the general public
    - ✓ Press release
    - ✓ Thematic events (such as health fairs and other health promotion programmes)

Evaluation of community diagnosis

- It is advisable to carry out evaluation after completing the community diagnosis.
- The following are the importance of evaluating community diagnosis process:
  - Identification of what diagnostic approaches that worked best versus those never worked
  - It provides answers to the following questions: Did the diagnosis reveal the hidden problem in the community? Was the process interfered with by biasness? Were there missing responses in the questionnaires? Was the correct sample size used? What is the accuracy level of the responses? What can be improved in future diagnosis processes?

Principles of community based research

- Community based research is a research that strives to be:
  - ✓ *Community situated* -begins with a research topic of practical relevance to the community (as opposed to individual scholars) and is carried out in community settings.
  - ✓ *Collaborative* -community members and researchers equitably share control of the research agenda through active and reciprocal involvement in the research design, implementation and dissemination.
  - ✓ *Action-oriented* -the process and results are useful to community members in making positive social change and to promote social equity.
- The following are its principles:
  - ✓ Recognizes community as a unit of identity
  - ✓ Builds on strengths and resources within the community
  - ✓ Facilitates collaborative partnerships in all phases of the research
  - ✓ Integrates knowledge and action for the mutual benefit of all partners
  - ✓ Promotes a co-learning and empowering process that attends to social inequalities and different perspectives
  - ✓ Disseminates findings and knowledge gains to all partners in an accessible way.
  - ✓ Involves long term commitment by all partners
  - ✓ The relevance of the research topic is identified or verified by community members.
  - ✓ The resources of research (financial, expertise, etc.) are shared with community members, particularly those most affected by the research topic.
  - ✓ The research process recognizes and addresses power imbalances between researchers and community members.

- ✓ The research process is driven by values, including: empowerment, supportive relationships, social change, learning as an ongoing process and respect for diversity.
- ✓ The research process and results are accessible and understandable to community members.
- ✓ The research process and results consider and adapt to the context in which the research is conducted.
- ✓ The research leaves a legacy, both in terms of the utilization of research results, as well as in the future collaboration among partners.

Roles of households and communities

- What are the roles of the following in community diagnosis? **{Give as group work}**
  - Individual ; Family ; Institutions; Faith based organization; Leaders

Participatory rural appraisal

- Is a technique typically used in the field to gather qualitative data, often to complement quantitative data
- Its **aims** are:
  - ✓ To involve local people in the selection, design, planning and implementation of programmes and projects that will affect them
  - ✓ Ensures that local perception, attitudes, values and knowledge are taken into account as fully as possible.
  - ✓ To make continuous and comprehensive feedback an integral part of all development activities.
  - ✓ It acts as alternative to the two common qualitative methods:
    - Questionnaires which often proved lengthy, costly and prone to errors,
    - Rushed site visits by researchers to collect haphazard data from local elites.
    - PRA uses a combination of approaches and methods to enable rural people to share enhance and analyze their knowledge of life and conditions, to plan and to act.
  - ✓ Uses a range of methods to ensure reliability and validity, and to enable cross-checking
  - ✓ Facilitate rapid and progressive learning with and from rural people
  - ✓ Enables vulnerable groups in a community to have a voice and impart their views on issues of transportation and access from which they are most often excluded.
  - ✓ Include: women, the elderly, disabled and even school children, researchers and other professionals are able to paint a realistic picture of community life; and through use of the different PRA techniques,
  - ✓ It answers the following questions relevant to community needs:
    - Which community members require the use of transport?
    - When is access required?
    - Why is access required?
    - Where to local people travel to?
    - How is transportation paid for?
    - What modes of transport to people take?

- PRA methods are based on the following **principles**:
  - ✓ A reversal of learning - to learn with and from rural people, directly, on the site, face to face, gaining from local, physical, technical and social knowledge.
  - ✓ Learning rapidly and progressively - with flexible use of methods, improvisation, iteration, and cross-checking, being adaptable in a learning process.

- ✓ Seeking diversity - Looking for, noticing and investigating contradictions, anomalies and difference.
- ✓ Triangulating – use of combined approaches for data collection
- ✓ Facilitating by the local people - Facilitating, investigation, analysis, presentation and learning by rural people themselves, so that they present and own their own outcomes.
- ✓ Preparation - Ensure that all available secondary data on the locality and subject has been reviewed, allowing suitable villages to be identified to capture a broad sample, before surveying commences.
- ✓ Facilitation - It is imperative that the external professional displays good facilitation skills, which aims to enable local people to undertake some or all of the investigation, mapping, modeling, diagramming, ranking, scoring, quantification, analysis, presentation and planning themselves.
- ✓ Behavior and attitudes - The behavior and attitudes of external facilitators are of primary importance, more important than methods even. All important attitudes include: critical self-awareness and embracing error, sitting down, listening and learning, not lecturing but allowing the villagers to be the main teachers and analysts.
- ✓ Longevity - Participatory approaches are not substitutes for, but are rather an integral part of, long term dialogue and sustained interaction. A single, brief participatory exercise with a group of local people will not lead to positive and lasting change.

- The following are **methods** of PRA
  - Triangulation
    - ✓ Triangulation is simply a method of linking different survey methods in order to crosscheck the information collected from each method.
    - ✓ Triangulation is of particular use when employing mapping, ranking and scoring, flow diagrams, Venn diagrams and wealth ranking techniques, as these involve group participation which may require some verification.

  - Observation
    - ✓ Prior to conducting any PRA techniques, researchers should be clear in their mind as to what exactly it is they are researching, and to have some realistic objectives of the PRA surveys.
    - ✓ Researchers should act on what they see, and recognize distinctions in gender, age, and wealth etc. amongst the community observation also aids improvisation, particularly when carrying out diagramming techniques, allowing household implements for example to be used in the PRA methods.

  - Semi-structured interviews
    - ✓ Participatory methods have contributed to adjusting the interview to make it more conversational, while still controlled and structured.
    - ✓ This is whereby only some of the questions and topics are predetermined, whilst the majority of questions will be formulated during the interview.
    - ✓ Questions are asked according to a flexible checklist and not from a formal questionnaire.

  - Ranking and scoring techniques
    - ✓ Ranking and scoring have long been used to assess people's expectations, beliefs, attitudes, preferences and opinions.
    - ✓ Ranking and scoring means placing something in order:
      - Ranking: putting in order
      - Scoring: weighting differences

- Matrix ranking
  - ✓ Matrices can be used to express other issues, where a two dimensional comparison is useful, such as where frequency or occurrence can be illustrated in the same way as preference scoring.
  - ✓ Matrix ranking involves listing the elements down one side, and the criteria on which they are judged, gained from informal discussion or pairwise ranking, across the top.
  - ✓ Each element is then considered in terms of each criterion.

- Diagramming
  - ✓ Diagrams, including maps, sketches and transects, summaries data in such a way that they can be used for different purposes such as planning, field discussion, and analysis and problem identification.
  - ✓ They are useful for opening up discussions between community members and the external team and helping clarify issues and questions.

- Participatory mapping
  - ✓ Maps can be used to identify the comparative location and importance of different resources within an area.
  - ✓ They can examine a great breadth of subject matter, and allow for a range of different types of map to be produced for one area or for comparative analysis by different groups within the same area.
  - ✓ Maps are also a useful tool to aid communication and to generate the following:
    - Provide a framework for discussion over the relative location of resources
    - Highlight resources of importance, using maps as a spatial guide
    - Raise issues which affect or are affected by these resources
    - Analyze the present status or condition of a location
    - Create a focus for interest in a discussion over resources
    - Social maps can be used to locate houses, services and infrastructure within an area.
    - Maps should be used as a visual stimulant, to identify the parameters faced by local people and to facilitate discussion about the importance people place on infrastructure provision etc.

- The following are **constraints** of PRA
  - The process of carrying out PRA methods requires mutual respect and trust between external facilitators and village participants; this takes time, and requires an understanding of the local culture.
  - Time consuming
  - A common mistake is, projects do not contact local government officials from the start - hence no local government support which is a prerequisite for success.
  - The step from problem identification with participatory PRA methods to follow up project definition is often poorly prepared.
  - Project officers should leave any class and gender biases they may carry at home!
  - The researcher must be able to listen but also to establish dialogue and share own experience.
  - Cultural protocol may be difficult to follow
  - It raises expectations

- A researcher stays in the village throughout the training session

Community Health Survey

Goal: Increase skills and knowledge on concepts and procedures of conducting community survey

Specific objectives:

- Understand the terms used in community health survey, reasons and types of survey
- Describe the process of carrying a survey, biased sample and response rate in relation to survey
- Outline points to note in constructing survey tool, causes of inaccuracy and report presentation
- Describe evaluation and feedback in relation to community survey

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Definitions

- **A survey** - is defined as a brief interview or discussion with individuals about a specific topic
- **Community survey** - is a way of asking group or community members what they see as the most important needs of that group or community.
- **Questionnaire** - is a research instrument consisting of a series of questions and other prompts for the purpose of gathering information from respondents.
- **Evaluation -** is a systematic determination of a subject's merit, worth and significance, using criteria governed by a set of standards.

Reasons – The following are reasons for carrying out a survey:

- Gather information about residents, their opinion, attitude and knowledge
- Measuring behaviors and population characteristics
- Solicit community reaction to policies, proposals and solutions
- Residents assess effectiveness of program, facilities and services
- Make residence aware of problems and their effects
- Provide residence access to policy making process
- Provide opportunity for communities to influence public decisions

When can you do a community survey?

- When your group is just starting out
- When there is doubt as to what the most important needs are
- When your group members disagree on this point among themselves
- When you need to convince outside funders or supporters that you are addressing the most important community problems
- When the community asks you to do it
- When you want to be sure that you will have community support for whatever you choose to do.

Types - Each survey is classified as either Anonymous of Non-anonymous

- An anonymous survey
  - ✓ Is used to get feedback from the users allowing them to give prompt feedback anonymously.
  - ✓ No user identifiable data will be recorded for the responses they will provide including usernames, identification particulars address etc.

- A non-anonymous survey
  - ✓ Allow you as a survey administrator to know who is giving feedback and what feedback he/she is giving.
  - ✓ In this survey type username (if user is logged in), identification particulars address, country details are recorded.

Sometimes, a survey can be divided into questionnaire and the interview.

- Questionnaires- are usually paper-and-pencil instruments that the respondent completes.
- Interviews - are completed by the interviewer based on the respondent says

Process – Good surveys start with clear theory, then the translation of theory into sensible survey questions, pre-testing and revision of questions, development and pre-testing of the survey instrument, sampling, data gathering, and, finally, data analysis.

The following is the process of planning and conducting a survey:

- Decide on your goals
  - ✓ Before you can start your research, you will need to form a clear picture in your mind of the expected outcome.
  - ✓ Identify the overall reasons for conducting the survey.
  - ✓ What will you know after you have conducted the survey? Do you need feedback on a product or your service? Is the information you are looking for of a general nature or very specific? Do you have a particular audience in mind, or will you be sending out online surveys to the general public?
  - ✓ The answers to these questions will help you to decide how to target your survey.
  - ✓ Determine the deadline for reporting the data the survey will collect. This will ensure you complete the survey on time

- Design the methodology for conducting the survey
  - ✓ Decide the procedures for conducting the survey: the number of people you will survey (sample size), how you will survey them (self-administered questionnaire or interview)

- Create a list of questions
  - ✓ Identify the questions you want the data to answer.
  - ✓ There are many different types of questions that can be used on a survey, like open questions, closed questions, matrix table questions, and single- or multi-response questions.
  - ✓ One of the benefits of designing an online survey is that participants don't have to fill in questions that are not relevant to them. Based on their answers certain questions can be skipped.
  - ✓ Develop the tools and pretest them

- Invite the participants
  - ✓ Who you want to take part in your survey will help you to decide on the best contact method.
  - ✓ Identify the study respondents and enumerators
  - ✓ You can send an email or call the participants
  - ✓ Train them and assign them responsibilities

- Gather your responses
  - ✓ Determine when to start the data collection
  - ✓ It is important to assign a unique code to each participant to make sure that they only take part in your survey once.
  - ✓ Distribute survey forms. As they are returned, track the number completed.

- Analyze the results
  - ✓ Analyze your date according to your objectives.
  - ✓ Make visual representations of the data by presenting the results in tables and graphs.

- ✓ You can also print out the data in the form of a spreadsheet, which can then be used for further analysis.
- ✓ There are several software packages available for data analysis (Epi-info, SPSS, SAS, STATA, RA)
- ✓ Present the analyzed report in Narratives, Graphs, Charts, Tables or Pictures

- Write a report
  - ✓ The final step in conducting online surveys is to write a report explaining your findings.
  - ✓ A successful survey will provide the answers to the questions you had about your business, product or service.
  - ✓ Use proper report writing layout
  - ✓ Provide summary of your results

Biased samples

- There are errors that may lead to inaccuracy when completing survey tools
- Biased sample means a sample has some favors on a particular subject.
- The following are causes of biasness:
  - Unrepresentative sample - Bias often occurs when the survey sample does not accurately represent the population. It is also called selection bias.
  - Under coverage - occurs when some members of the population are inadequately represented in the sample
  - Nonresponse bias - Sometimes, individuals chosen for the sample are unwilling or unable to participate in the survey.
  - Voluntary response bias - Voluntary response bias occurs when sample members are self-selected volunteers, as in voluntary samples
  - Measurement Error- A poor measurement process can also lead to bias
  - Response bias - refers to the bias that results from problems in the response process. Some examples of response bias are given below:
    - *Leading questions* - The wording of the question may be loaded in some way to unduly favor one response over another.
    - *Social desirability* - Most people like to present themselves in a favorable light, so they will be reluctant to admit to unsavory attitudes or illegal activities in a survey, particularly if survey results are not confidential. Instead, their responses may be biased toward what they believe is socially desirable.

- Biased sample can be eliminated by applying random sampling procedures.
- Random sampling is a procedure for sampling from a population in which:
  - The selection of a sample unit is based on chance
  - Every element of the population has a known, non-zero probability of being selected.

Response rate

- Response rate - refers to the number of people who answered the survey divided by the number of people in the sample. It is also known as completion rate or return rate. It is usually expressed in the form of a percentage.
- The following are ways of increasing response rate:
  - ✓ Offering an incentive to the participant, for example, you can offer a gift, the chance of winning something in a lottery, a donation to charity, or a point's accumulation system where participant can save up points that can be exchanged for gifts.
  - ✓ Promising to share the results with your participants.

Points to be considered when constructing a survey tool

- Open ended questions with probes
- Simple and brief questions
- Avoid closed questions
- Avoid leading questions
- Start with most easy questions
- Make personal questions last
- Each question to address one issue
- When writing the questions, keep the language very simple and avoid ambiguity or double negations.
- Make questions specific
- Ask questions in a logical order
- Construct response categories carefully – let it not be too long
- Provide clear and sufficient instructions or directions including reasons for the survey

Errors leading to inaccuracy when constructing a survey tool

- Unclear objectives
- Questions not matching the survey objectives
- Questions are ambiguous
- Tool doesn't provide adequate space for data entry
- Questions are not well numbered

Survey report presentation

- Every survey must have a report. This can be a brief summary (abstract) of detailed report
- Abstract- provides a one paged summary of the survey report in terms of:
    - Title, authors particulars and introduction
    - Methodology
    - Key findings
    - Recommendations
- A detailed report is voluminous and it contains the following:
    - Title page (table of content, acknowledgements, list of abbreviations, executive summary/ preface or abstract)
    - Chapter1 – Background and introduction
    - Chapter2 – Literature review
    - Chapter3 – Methodology/data collection process
    - Chapter4 – Findings/data results and interpretations
    - Chapter5 – Discussions of the findings using your own thoughts and opinion
    - Chapter6 – Conclusions and recommendations – this can also include detailed implementation plan
    - Reference

Evaluation and feedback

- Evaluation – is an appraisal of something to determine its worth or fitness.
- It can be done before starting a project (Formative) or after completion of the project (Summative)
- The reasons for evaluation are:

- It can assist an organization, program, project or any other intervention or initiative to assess any aim, realizable concept/proposal, or any alternative,
- To help in decision-making;
- To ascertain the degree of achievement or value in regard to the aim and objectives and results of any such action that has been completed.[
- Is to enable reflection and assist in the identification of future change
- Determines if specific program strategies were implemented as planned
- Focuses on program implementation
- Helps to draw conclusions about five main aspects of the intervention:
  - ✓ Relevance
  - ✓ Effectiveness
  - ✓ Efficiency - Finally, cost-benefit or cost-effectiveness analysis assesses the efficiency of a program. Evaluators outline the benefits and cost of the program for comparison. An efficient program has a lower cost-benefit ratio.
  - ✓ Impact
  - ✓ Sustainability

- Program evaluations can involve both **quantitative** and **qualitative** methods of social research
- Monitoring is a systematic and routine collection of information from projects and programmes for four main purposes:
  - To learn from experiences to improve practices and activities in the future;
  - To have internal and external accountability of the resources used and the results obtained;
  - To take informed decisions on the on-running actions
- Monitoring and evaluation uses **indicators** to for measurement and in drawing conclusions
- Indicators are statistics used to measure current conditions as well as forecasting or predicting trends. They measure change over time. Indicators can be:
  - Input – measures efforts put in
  - Process – measures procedures
  - Output – measures immediate results
  - Outcome – measures changes that comes as a result of the outputs
- Monitoring and evaluation is implemented through a logical framework (commonly known as log-frame)
- The monitoring and evaluation log frame include:
  - Objectives of the program
  - Activities implemented
  - Measurable indicators
  - Means of verification

M&E Log frame

Objective	**Activity**	**Indicators**	**Means of verification**
To reduce the number of	1.Distribute Chlorine to every homes	# of homes supplied with chlorine	Household survey

deaths due to diarrheal diseases from 20 per month to 15 per month by 2015		# of homes with patients with diarrhea # of chlorines distributed per home	Household survey Distribution list
	2.Educate family members in 20 homes in the use of water treatment	# of homes covered with education on use of water treatment # of family members educated on use of water treatment # of educated family members actually treating water	Education program Education report Home visit
	3.Drill 2 boreholes in the community	# of boreholes drilled in the community # of drilled boreholes has water # of people fetching water from the borehole daily	Survey Survey Survey
	4.Construct pit latrines in 30 homesteads	# of latrines constructed # of homes with latrines # of constructed latrines are in use	Construction report Household survey Household survey
	5.Provide ORS to children with diarrhea	# or ORS sachets provided # of children provided with ORS	S11 form ORS register

- In any survey, there must be feedback
- A survey feedback is a process in which community members complete questionnaires on various issues, receive feedback on the results, and then take appropriate actions to address the critical needs and concerns.
- The objectives of survey feedback are:
    - To assist the organization in diagnosing its problems and developing action plan for problem-solving.
    - To assist the group members to improve the relationships through discussion of common problems.
- The following are the processes of survey feedback:
    - Data Collection - The first step in survey feedback is data collection usually by a consultant based on a structured questionnaire.
    - Feedback of Information - After the data are analyzed, feedback is given to the persons who have participated in the fulfilling up of questionnaire. The feedback may be given

either orally or in a written form. In oral system of feedback, it is provided through group discussion or problem-solving sessions conducted by the consultant.

- Follow-up Action - Survey feedback programme is not meaningful unless some follow-up action is taken based on the data collected. One such follow-up action may be to advise the participants to develop their own action plans to overcome the problems revealed by the consultant.

Community strategy

Goal: Improve understanding on community strategy

Specific objectives:

- To understand the meaning and objectives of community strategy
- To define the household and community roles in community strategy
- To understand the implementation process of community strategy
- Improve knowledge on community involvement and participation in sustaining health programs
- To gain practical experience on community units

Community strategy - this is an approach that sets up an assurance that Kenyan communities have the capacity and motivation to take up their essential role in health care delivery

National Health Sector Strategic Plan – This is a five year plan that aims at reversing the declining trends in key health sector indicators. The NHSSP II has five broad policy objectives:

- Increase equitable access to health services
- Improve the quality and responsiveness of services in the sector
- Improve the efficiency and effectiveness of service delivery
- Enhance the regulatory capacity of MOH
- Foster partnerships in improving health and delivering services
- Improve the financing of the health sector

Kenya Essential Health Package of Health - KEPH is an approach that supports the achievement of the goals of the second National Health Sector Strategic Plan, 2005–2010 (NHSSP II). The packages are:

- Eliminate communicable conditions
- Revise rising burden of non-communicable conditions
- Reduce the burden of violence and injuries
- Provide essential health services
- Minimize exposure of health risk factors
- Strengthen collaboration with other related health sectors

Household – It consists of one or more people who live in the same dwelling and also share at meals or living accommodation, and may consist of a single family or some other grouping of people. A single dwelling will be considered to contain multiple households if either meals or living space are not shared.

Community mobilization – is an attempt to bring both human and non-human resources together to undertake developmental activities in order to achieve sustainable development.

Community involvement - is the process of engaging in dialogue and collaboration with community members

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Goal- is a desired result that a person or a system envisions, plans and commits to achieve. The overall goal of the community strategy is to enhance community access to health care in order to improve productivity and thus reduce poverty, hunger, and child and maternal deaths, as well as improve education performance across all the stages of the life cycle.

Objectives – the objectives of community strategy are as follows:

- To improve the health status of Kenyan communities through initiation and implementation of life-cycle focused health actions.
- Provide level 1 services for all cohorts and socio-economic groups taking in to account their priority needs
- Build the capacity of community health extension workers (CHEWs) and community own resource persons (CORPS) to provide services at level 1
- Strengthen health facility- community linkages through effective decentralization and partnership
- Strengthen the community to progressively realize their rights for accessible and quality care and to seek accountability from facility based health services

How it came about

- Communities are at the foundation of affordable, equitable and effective health care, and are the core of Kenya Essential Package for Health (KEPH)
- The concept came about after seeing the need to enhance community access to health care in order to improve productivity and reduce poverty, hunger and child & maternal dearth as well as improve education performance across all stages of the life cycle
- Kenya realized the need of establishing sustainable and decentralized community level services and accountability

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Roles of households and community

- Health promotion – through:
    - Ensure healthy diet for people at all stages in life in order to meet nutritional need
    - Building healthy social capital to ensure mutual support in meeting daily needs
    - Demand health and social entitlements as citizens
    - Monitor health status to promote early detection of problem for timely action
    - Taking regular exercise
    - Ensure gender equity
    - Use available services to monitor nutrition, chronic conditions and other causes of disability

- Disease prevention – through:
    - Practice good personal hygiene in terms of hand washing, use of latrine etc
    - Using safe drinking water
    - Ensure adequate shelter and protection against vectors of diseases
    - Preventing accidents and abuse and taking appropriate measures when they occur
    - Ensuring appropriate sexual behavior to prevent STI

- Care seeking compliance with treatment and advice –through:
    - Giving sick household members appropriate home care for illness
    - Taking children for immunization as scheduled
    - Recognizing and acting on the need for referral or seeking care outside home
    - Following recommendations given by health worker in relation to treatment, follow up and referral
    - Ensuring that every pregnant woman receives antenatal and maternity care services

- Governance and management of health services – through:
    - Attending and taking an active part in meetings to discuss:
        - Trends in coverage

        - Morbidity
        - Resources and clients satisfaction

- Claiming rights – through:
    - Knowing what rights communities have in health
    - Building capacity to claim these rights progressively
    - Ensuring that health workers in the community are accountable for effective health service delivery and resource use
    - Ensure health facilities are functioning in line with the Citizen Health Charter

Implementation framework and process

- Organizational structure of coordination

Structure	Summary of activities
Household, village, school, congregation	• Plan, implement, monitor, evaluate and provide feedback on activities • Mobilize and manage resources • Undertake health promotion, hygiene, lifestyle and care seeking initiatives
Sub location, parish, dispensary	• Provide health status information • Support, provide and protect water • Manage CBHIS • Train CORPS Establish and maintain linkages
Division, health centre	• Provide training and support supervision • Coordinate, collaborate, network, pool resources
District, diocese, hospital	• Comprehensive district planning • Strengthen health boards • Facilitate M&E and coordinate NGOs • Train leaders and extension staff
Province	• Technical support • Build capacity of districts
National	• Formulate guidelines and policy • Build capacity of districts and province • Multi-sector and donor coordination • Ensure CBHIS is part of the HMIS • Quality assurance

- The entry steps
    - Define a clear implementation guideline
    - Create awareness among district leaders
    - Form and equip district-level multi-sector working groups and training teams
    - Follow up, monitor and evaluate
    - Launch the program in the communities. This will follow steps like:
        - Exploration
        - Protocol
        - Participatory assessment

    - The assessment should include:
        - The population
        - Community structures
        - Any existing community information systems

        - Resource availability
        - Communication strategy
        - Service delivery and the minimum package of care and support
        - Health status
        - Food security
        - The environment

- Feedback and participatory planning of level 1 service activities
    - Findings of the assessment are presented to the community in a general meeting
    - The community validates the findings
    - The community summarize the findings
    - Strategic action points are based on strengths and opportunities
    - Actions are categorized in to :
        - Those that can be taken by the CORPS
        - Those that require additional support from local partners
        - Those that require external support

Community mobilization -This is the process of bringing the community members together. It is is an attempt to bring both human and non-human resources together to undertake developmental activities in order to achieve sustainable development

- is a process through which action is stimulated by a community itself that is planned, carried out, and evaluated by a community's individuals, groups, and organizations on a participatory and sustained basis to improve the health, hygiene and education levels so as to enhance the overall standard of living in the community
- is a process which begins a dialogue among members of the community to determine who, what, and how issues are decided,
- The process of sensitizing and motivating the community to work together
- Effective mobilization will ensure the community's interest is created and community members are motivated and influenced to take action and participate
- Mobilization is done through community gatherings, baraza, village health days, women and youth groups

Steps of community mobilization

- Getting to know the community, provide information & create interest
- Analysis + Identification resources + Problem identification + Identification solutions & projects
- Decision making & selection of leaders by community
- Identification of stakeholders
- Planning & selection of implementing leader
- Project implementation by community
- Follow-up & monitoring
- Evaluation

Tasks in community mobilization

- Developing an on-going dialogue between community members
- Creating or strengthening community organizations (Committees etc.)
- Creating an environment in which individuals can empower themselves to address their own and their community's needs
- Promoting community members' participation

Community involvement and participation - Community involvement is the process of letting the community members play active role in any activity. A project is more likely to fail if there is no community 'buy in' - Sometimes this is because the project was introduced by 'experts' who had their own ideas about what should be done. This is called the top-down approach

- Importance of community involvement and participation
  - ✓ It promote ownership and accountability
  - ✓ CORPS learn much faster when they participate
  - ✓ Community can participate in health promotion, advocacy, follow up
  - ✓ It is often better to use a participatory approach to ensure the project's sustainability
  - ✓ The participation helps to ensure that programmes are equitable and effective.
  - ✓ Participation can positively impact the health, well-being and safety.
  - ✓ Communities have different needs, problems, beliefs, practices, assets, and resources related to sexual health. Getting the community involved in program design and implementation helps ensure that strategies are appropriate for and acceptable to the community and its youth.
  - ✓ Community participation promotes shared responsibility by service providers, community members, and youth themselves for the sexual health of adolescents in the community.
  - ✓ Community participation can increase the accountability of sexual health programs and service providers.
  - ✓ Participation can empower youth within the community

- Key messages to community involvement and participation
  - ✓ Special attention should be given to ensuring that all groups are able to participate, including those with specific needs and/or those who are marginalized and lacking a voice in decision-making.
  - ✓ Participation structures and procedures may require targeted effort and additional resources; it is an essential way of reinforcing a sense of dignity, reducing vulnerability and helping to build local capacity without undermining peoples' own coping strategies in times of crisis.
  - ✓ Participation should be based on assessments of existing structures which can be used to support participatory methods.
  - ✓ Participatory approaches should respect local culture there needs to be due regard for culturally-embedded power relationships which may be exploitative or oppressive.
  - ✓ Participation and community involvement should be planned and integrated into all stages of the project cycle – assessment, planning, implementation, monitoring and evaluation
  - ✓ A community's members are a rich source of knowledge about their community and of energy and commitment to that community

Procedures of community based research

- Purposes of community based research:
  - o There is increased focus of funding agencies and researchers on approaches to public health research
  - o Influence of social and political systems on behavioral and health outcomes

- Principles of community-based research:
  - ✓ Respect for community interests that go beyond those of individual potential research subjects
  - ✓ Identification of potential community stakeholders as well as individual research subjects.
  - ✓ Inform the community stakeholders and potential research subjects about the research.
  - ✓ Invite feedback regarding concerns about the research from community stakeholders and individual potential research subjects.

- ✓ Partnership establishment
- ✓ Respect the community partner's interest in the project and be open to ways that the community might benefit or want to use the information.
- ✓ Disseminate research findings to both community stakeholders and individual research subjects.
- ✓ Having the research topic address a community-defined need, question or problem, and strive to combine knowledge with action to achieve necessary changes.
- ✓ Recognizing the research as a partnership and be open to the guidance of community insight and experience.
- ✓ Having the partnership appropriately balance power and decision-making between the researchers and the community participants in a way that is mutually acceptable.
- ✓ Partners making clear and open communication an ongoing priority by striving to understand each other's needs and self-interests.
- ✓ Partners recognizing race, ethnicity, class, and other aspects of culture matter and talk openly about these issues.
- ✓ Feedback among all stakeholders in the partnership, with the goals of continuously improving the partnership and its outcomes.
- ✓ Realizing partnerships can dissolve and plan a process for closure

Income generating activities

- This is a mechanism for sustainability
- The CORPS get involved in activities that give them revolving funds
- Some of the IGAs include kitchen gardening, poultry raring, merry-go-round (chama)

Skills in training a community resource person

- Adult learn best through illustrated lectures, discussions, brainstorming, group work and feedback, role plays, demonstration, community or household visit, transect walk in PRA, observation and resource mapping.
- Adult learning theory: *problem-based and collaborative*
- Principles of adult learning:
  - *Adults are internally motivated and self-directed*
    - Adult learners resist learning when they feel others are imposing information, ideas or actions on them.
    - A facilitator should:
      - Set up a *graded learning program* and do not overloading for the student.
      - *Develop rapport* with the student and show interest in their opinion
      - *Lead the student toward inquiry* before supplying them with too many facts.
      - Provide *regular constructive and specific feedback*
      - *Review goals* and *acknowledge goal completion*
      - *Encourage use of learning resources*
      - *Acknowledge the preferred learning style* of the student.

  - *Adults bring life experiences and knowledge to learning experiences*
    - ✓ Adults like to be given opportunity to use their existing knowledge and experience gained from life experience, and applies it to their new learning experiences.
    - ✓ *Find out about your student* - their interests and past experiences
    - ✓ *Assist them to draw on those experiences*
    - ✓ *Facilitate reflective learning opportunities*

- *Adults are goal oriented*
  - ✓ Adult students become ready to learn when "they experience a need to learn it in order to cope more satisfyingly with real-life tasks or problems"
  - ✓ As educator, you can:
    - *Provide meaningful learning experiences*
    - *Provide real case-studies*
    - *Ask questions* that motivate reflection, inquiry and further research.

- *Adults are relevancy oriented*
  - ✓ Adult learners want to know the relevance of what they are learning to what they want to achieve.
  - ✓ Ways of helping learners see the value of their observations and practical experiences are:
    - *Ask the student to do some reflection*
    - *Provide some choice* of fieldwork project by providing two or more options, so that learning is more likely to reflect the student's interests.

- *Adults are practical*
  - ✓ Students move from classroom mode to hands-on problem solving
  - ✓ A trainer should:
    - *Clearly explain your clinical reasoning*.
    - *Be explicit*.
    - *Promote active participation*

- *Adult learners like to be respected*
  - ✓ Respect can be demonstrated to your student by:
    - *Taking interest*
    - *Acknowledging the wealth of experiences* that the student brings to the placement;
    - *Regarding them as a colleague* who is equal in life experience
    - *Encouraging expression* of ideas, reasoning and feedback at every opportunity.

Environment and Health

Goal: Understand the role of environment in our health

Specific objectives:

- Describe environmental factors that affect our health
- Describe the relationship of man with the environment

Environment – is anything that externally surrounds a man. This is where man lives. The World Health Organization (WHO) defines environment, as it relates to health, as "all the physical, chemical, and biological factors external to a person, and all the related behaviors."

Environmental health – is the relationship between environment and health

Belief – is an acceptance that a statement is true or that something exists. It can also be defined as an acceptance that something exists or is true, especially one without proof

Culture - Culture refers to the cumulative deposit of knowledge, experience, beliefs, values, attitudes, meanings, hierarchies, religion, notions of time, roles, spatial relations, concepts of the universe, and material objects and possessions acquired by a group of people in the course of generations through individual and group striving

Environmental components – these are what makes up the environment. These include:

- PHYSICAL: - e.g. air, water, soil, housing, climate, geography, heat, light, noise, debris, radiation
- BIOLOGICAL – e.g. Man, viruses, microbial agents, insects, rodents, animals and plants, etc.
- PSYCHOSOCIAL – e.g. Cultural values, customs, beliefs, habits, attitudes, morals, religion, education, lifestyles, community life, health services, social and political organization.

Importance of environmental health

- Goal of environmental health is to promote health for all through a healthy environment.
- Environmental health consists of preventing or controlling disease, injury, and disability related to the interactions between people and their environment.
- It increasing quality of life and years of healthy life
- Globally, nearly 25 percent of all deaths and the total disease burden can be attributed to environmental factors such as:
    - Exposure to hazardous substances in the air, water, soil, and food
    - Natural and technological disasters
    - Emergencies
    - Water supply
    - Increased urban development
    - Physical hazards
    - Nutritional deficiencies
- Poor environmental quality has its greatest impact on people whose health status is already at risk.
- Therefore, environmental health must address the societal and environmental factors that increase the likelihood of exposure and disease.

Environmental health factors (risks) affecting our health

- Environmental health risks are factors outside of the body that can affect a person's wellbeing.
- Examples include:

    - Air pollution – for example, smog, wood smoke and mould
    - Water quality – for example, grey water, tank water, fluoridation and drought.
    - Food quality and safety – for example, contamination and nutrition.
    - Chemicals – for example, pesticides, farm chemicals.
    - Metals – for example, exposure to lead, mercury
    - Diseases from animals and insects (vector borne) – for example, dengue fever, virus, malaria.
    - Infectious diseases – for example, viral infections like swine flu.
    - Natural hazards – for example, solar radiation and extreme weather events.
    - Man-made structures – for example, exposure to asbestos or electromagnetic radiation sources like mobile phones.
    - Occupational health – for example, safety issues relating to the workplace such as noise pollution and hazardous waste.
    - Climate change – for example, higher sea levels, increased soil salinity and increased temperatures.
    - Housing conditions
- Preventing or reducing the risk of illness, injury or disease in the community is important for environmental health.

Understanding Environmental Health

- The following are 6 themes that draw attention to elements of the environment and their linkages to health.

Outdoor Air Quality

- Poor air quality is linked to premature death, cancer, and long-term damage to respiratory and cardiovascular systems.
- Decreasing air pollution is an important step in creating a healthy environment.

Surface and Ground Water

- Surface and ground water quality applies to both drinking water and recreational waters.
- Contamination by infectious agents or chemicals can cause mild to severe illness.
- Protecting water sources and minimizing exposure to contaminated water sources are important parts of environmental health.

Toxic Substances and Hazardous Wastes

- The health effects of toxic substances and hazardous wastes are not yet fully understood. Research to better understand how these exposures may impact health is ongoing.
- Meanwhile, efforts to reduce exposures continue.

- Reducing exposure to toxic substances and hazardous wastes

Homes and Communities

- People spend most of their time at home, work, or school. Some of these environments may expose people to:
  - Indoor air pollution
  - Inadequate heating and sanitation
  - Structural problems
  - Electrical and fire hazards
  - Lead-based paint hazards

Infrastructure and Surveillance

- Prevention of exposure to environmental hazards relies on many partners, including state and local health departments.
- Personnel, surveillance systems, and education are important resources for investigating and responding to disease, monitoring for hazards, and educating the public.
- Additional methods and greater capacity to measure and respond to environmental hazards are needed.

Global Environmental Health

- Water quality is an important global challenge.
- Diseases can be reduced by improving water quality and sanitation and increasing access to adequate water and sanitation facilities.

Relationship of individual, family, community and environment

- Humans interact with the environment constantly. These interactions affect quality of life, years of healthy life lived, and health disparities.
- Individuals group together to make a family. Families group together to make a community. Individuals, families and communities live in the environment
- Environment hosts the individual and members of the family and community

Responsibilities if health teams in environmental health

Physician

- Performing complete health assessments including: Taking a full medical history including presenting
- Complaint, past illnesses, social history, family history, and performing a complete physical examination
- Screening patients at risk for hereditable conditions and potentially preventable disorders
- Assessment, diagnosis, primary medical treatment and advice for management of acute medical conditions and injuries
- Assessment of the exacerbations and complications of chronic medical problems

Nurses

- Assess holistically and provide services to patients in all developmental stages, and to families and communities
- Complete health assessments, including a health history and physical examination
- Formulate and communicate medical diagnoses

- Synthesize information from patients to identify broader implications for health within the family
- Use family assessment tools to evaluate family strengths and needs
- Determine the need for, and order from, an approved list of screening and diagnostic laboratory tests and interpret the results
- Determine the need for, and order and interpret reports of X-rays, ECGs and diagnostic ultrasounds for diagnosis

Social worker

- Assessment of social problems by obtaining case history and background information
- Assessment of eligibility of social benefits

****Brainstorm roles of the following*****

- Health records and information officer
- Physiotherapy
- Occupational therapy
- Nutrition Officer
- Environmental health officer

How Culture Influences Health -Key points

- Culture is a pattern of ideas, customs and behaviors shared by a particular people or society. It is constantly evolving.
- The speed of cultural evolution varies. It increases when a group migrates to and incorporates components of a new culture into their culture of origin.
- Children often struggle with being 'between cultures'– balancing the 'old' and the 'new'. They essentially belong to both, whereas their parents often belong predominantly to the 'old' culture.
- The influence of culture on health is vast:
    - It affects perceptions of health, illness and death,
    - Beliefs about causes of disease
    - Approaches to health promotion
    - How illness and pain are experienced and expressed, where patients seek help
    - The types of treatment patients prefer.
- Both health professionals and patients are influenced by their respective cultures.
- Cultural bias may result in very different health-related preferences and perceptions.
- Demonstrating awareness of a patient's culture can promote trust, better health care, lead to higher rates of acceptance of diagnoses and improve treatment adherence.
- Acceptance of a diagnosis, including who should be told, when and how.
- Acceptance of preventive or health promotion measures (e.g., vaccines, prenatal care, birth control, screening tests, etc.).
- Perceptions of death, dying and who should be involved
- Willingness to discuss symptoms with a health care provider, or with an interpreter being present.
- Influence of family dynamics, including traditional gender roles, filial responsibilities, and patterns of support among family members.
- How accessible the health system is, as well as how well it functions.

How man has interfered with the environment

- Technology

- Agricultural land los and soil erosion
- Energy industry pollution
- Creation of infrastructure
- Coal mining and burning
- **Pharmaceuticals and personal care products**
- Creation of artificial ecosystem
- Nuclear power

Role of community in preserving Healthy and Safe Environments

What can state, local and territorial governments do?

- Facilitate collaboration among diverse sectors (e.g., planning, housing, transportation, energy, education, environmental regulation, agriculture, business associations, labor organizations, health and public health) when making decisions likely to have a significant effect on health.
- Include health criteria as a component of decision making (e.g., policy making, land use and transportation planning).
- Conduct comprehensive community health needs assessments and develop state and community health improvement plans.
- Strengthen and enforce housing and sanitary code requirements and ensure rapid remediation or alternative housing options.
- Participate in national voluntary accreditation of health departments.

What Can Businesses and Employers Do?

- Ensure that homes and workplaces are healthy, including eliminating safety hazards (e.g., trip hazards, unsafe stairs), ensuring that buildings are free of water intrusion, indoor environmental pollutants (e.g., radon, mold, tobacco smoke), and pests, and performing regular maintenance of heating and cooling systems.
- Adopt practices to increase physical activity and reduce pollution.

What Can Health Care Systems, Insurers, and Clinicians Do?

- Partner with state, local, and territorial governments, business leaders, and community-based organizations to conduct comprehensive community health needs assessments and develop community health improvement plans.
- Support integration of prevention and public health skills into health care professional training and cross train health care practitioners to implement prevention strategies.
- Increase the use of certified equipment and skilled personnel

What Can Early Learning Centers, Schools, Colleges, and Universities Do?

- Integrate appropriate core public health competencies into relevant curricula (e.g., nursing, medicine, dentistry, allied health, pharmacy, social work, education) and train professionals to collaborate across sectors to promote health and wellness.
- Implement policies and practices that promote healthy and safe environments (e.g., improving indoor air quality; addressing mold problems; reducing exposure to pesticides and lead; ensuring

that drinking water sources are free from bacteria and other toxins; implementing and enforcing tobacco free policies).

What Can Community, Non-Profit, and Faith-Based Organizations Do?

- Convene diverse partners and promote strong cross-sector participation in planning, implementing, and evaluating community health efforts.
- Implement processes to ensure that people are actively engaged in decisions that affect health.

What Can Individuals and Families Do?

- Use alternative transportation (e.g., biking, walking, public transportation, car and vanpooling).
- Conduct home assessments and modifications (e.g., installing smoke and carbon monoxide detectors, testing for lead, checking for mold and radon).
- Purchase energy-efficient products, support local vendors, and recycle.

Common barriers to inter-professional healthcare team work

Organizational barriers:

- Lack of knowledge and appreciation of the roles of other health professionals;
- Compelling arguments for team building to senior decision-makers;
- Lack of outcomes research on collaboration;
- Financial and regulatory constraints;
- Legal issues of scope of practice and liability;
- Reimbursement structures for different professions, including which services receive
- Hierarchical administrative and educational structures that discourage inter-professional collaboration.

Barriers at the team level:

- Lack of a clearly stated, shared, and measurable purpose;
- Lack of training in inter-professional collaboration;
- Role and leadership ambiguity;
- Team too large or too small;
- Team not composed of appropriate professionals;
- Lack of appropriate mechanism for timely exchange of information;
- Need for orientation for new members;
- Lack of framework for problem discovery and resolution;
- Difference in levels of authority, power, expertise, income;
- Difficulty in engaging the community;
- Traditions/professional cultures, particularly medicine's history of hierarchy;
- Lack of commitment of team members;
- Different goals of individual team members;
- Apathy of team members;
- Inadequate decision making; and
- Conflict regarding individual relationships to the patient/client.

Barriers faced by individual team members:

- Split loyalties between team and own discipline;
- Multiple responsibilities and job titles;
- Competition
- Gender, race, or class-based prejudice;

- Persistence of a defensive attitude;
- Reluctance to accept suggestions from team members representing other professions; and
- Lack of trust in the collaborative process.
- Barriers for independent providers:
- Accustomed to assuming total responsibility;
- Unease with allowing others to be involved in clinical decision-making;
- Discomfort with performance review by team members of different professional backgrounds;
- Legal liability for others' decisions; and
- Dilution of traditional one-to-one relationship with patient/client.

Overcoming barriers

- Agree on unifying philosophy centered around primary care of the patient/client and the community;
- Develop a commitment to the common goal of collaboration;
- Learn about other professions;
- Respect others' skills and knowledge;
- Establish positive attitudes about own profession;
- Develop trust between members;
- Be willing to share responsibility for patient/client care;
- Establish a mechanism for negotiation and re-negotiation of goals and roles over time;
- Establish method for resolving conflicts between team members; and
- Be willing to work continuously on overcoming barriers.

Drugs and substance abuse

Goal: Understand common drugs abused and ways of controlling drug abuse

Specific objectives:

- Discuss commonly abused drugs and its effects in our health
- Gain skills and knowledge on behavior change communication

Drug abuse – is the excessive use of a substance, especially alcohol or a drug. This is a compulsive, excessive, and self-damaging use of habit forming drugs or substances, leading to addiction or dependence, serious physiological injury (such as damage to kidneys, liver, heart) and/or psychological harm (such as dysfunctional behavior patterns, hallucinations, memory loss), or death. Also called substance abuse

Classifications of drugs

Drugs may be categorized or classified according to certain shared symptomatology or effects. Drugs from each of these categories can affect a person's central nervous system and impair a person's normal faculties, including a person's ability to safely operate a motor vehicle.

- *Central Nervous System (CNS) Depressants* - CNS Depressants slow down the operations of the brain and the body. Examples include alcohol, barbiturates, anti-anxiety tranquilizers (e.g., Valium, Librium, Xanax, Prozac, and Thorazine)

- *CNS Stimulants* - CNS Stimulants accelerate the heart rate and elevate the blood pressure and "speed-up" or over-stimulate the body. Examples include Cocaine, "Crack", Amphetamines and Methamphetamine ("Crank").

- *Hallucinogens* - Hallucinogens cause the user to perceive things differently than they actually are. Examples include LSD, Peyote, Psilocybin and MDMA (Ecstasy).

- *Dissociative Anesthetics* - It includes drugs that inhibit pain by cutting off or dissociating the brain's perception of the pain. PCP and its analogs are examples of Dissociative Anesthetics.

- *Narcotic Analgesics* - A narcotic analgesic relieves pain, induces euphoria and creates mood changes in the user. Examples of narcotic analgesics include Opium, Codeine, Heroin, Demerol, Darvon, Morphine, Methadone, Vicodin and OxyContin.

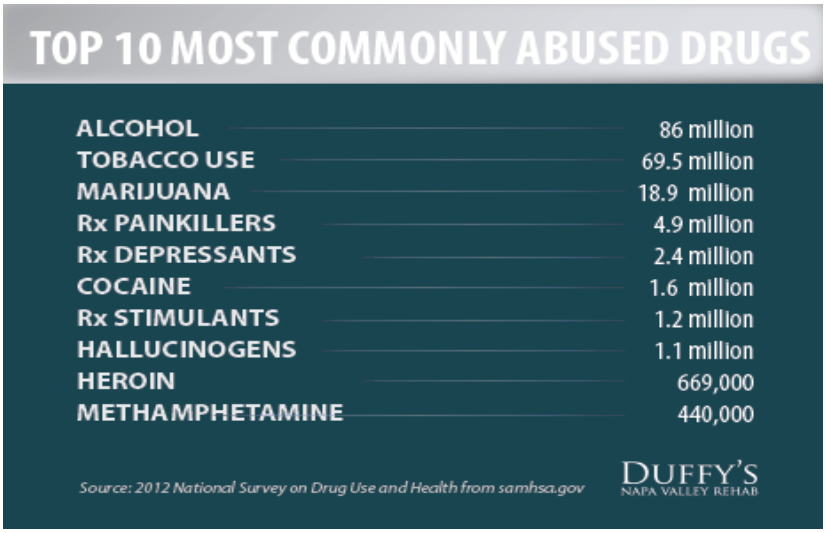
- *Inhalants* - Inhalants include a wide variety of breathable substances that produce mind-altering results and effects. Examples include Toluene, plastic cement, paint, gasoline, paint thinners, hair sprays and various anesthetic gases.

- *Cannabis* - Cannabis is the scientific name for marijuana. The active ingredient in cannabis is delta-9 tetrahydrocannabinol. This category includes cannabinoids and synthetics like Dronabinol.

Commonly abused drugs

- *Alcohol* - In chemistry, an alcohol is any organic compound in which the hydroxyl functional group is bound to a saturated carbon atom. The term alcohol originally referred to the primary alcohol ethyl alcohol, the predominant alcohol in alcoholic beverages
- *Tobacco* - is a product prepared from the leaves of the tobacco plant by curing them. The plant is categorized within the genus *Nicotiana* of the Solanaceae (nightshade) family

- *Marijuana* - Cannabis, also known as marijuana and by numerous other names, is a preparation of the Cannabis plant intended for use as a psychoactive drug or medicine
- *Pain killers* - An analgesic or painkiller is any member of the group of drugs used to achieve analgesia, relief from pain. Analgesic drugs act in various ways on the peripheral and central nervous systems
- *RX sedatives* - Prescription medications that act as central nervous system depressants. Barbiturates are prescription sedatives or "sleeping pills" Valium and benzodiazepines are prescription "tranquilizers.



- *RX stimulants* - Prescription stimulants were used historically to treat asthma, obesity, neurological disorders, and a variety of other ailments, before their potential for abuse and addiction became apparent
- *Cocaine*- Cocaine, also known as benzoylmethylecgonine, is a strong stimulant mostly used as a recreational drug. It is commonly snorted, inhaled, or injected into the veins
- Hallucinogens - A hallucinogen is a psychoactive agent which can cause hallucinations, perception anomalies, and other substantial subjective changes in thoughts, emotion, and consciousness. The common types of hallucinogens are psychedelics, dissociative, or deliriants
- *Heroin* - also known as diamorphine and commonly known by its street names of smack, boy, horse, brown, black, tar, and others is an analgesic originally synthesized by in 1874

*****Group work and presentation ******

Identify effects of drugs and substance abuse to: Individuals; Household and Community

Illnesses associated with drug abuse

- *HIV* - The human immunodeficiency virus (HIV) causes HIV infection and the acquired immunodeficiency syndrome (AIDS).
- *Depression* - A depressive disorder is a syndrome (group of symptoms) that reflects a sad, blue mood exceeding normal sadness or grief. Depressive disorders are characterized not only by negative thoughts, moods, and behaviors but also by specific changes in bodily functions (for example, eating, sleeping, and sexual activity).
- *Erectile dysfunction* - Erectile dysfunction (ED), also known as impotence, is the inability to achieve or sustain an erection for satisfactory sexual activity.
- *Schizophrenia* - Sometimes colloquially but inaccurately referred to as split personality disorder, schizophrenia is a chronic, severe, debilitating mental illness
- *Steroid withdrawal symptoms* - can mimic many other medical problems. Weakness, fatigue, decreased appetite, weight loss, nausea, vomiting, diarrhea(which can lead to fluid and electrolyte abnormalities), and abdominal pain are common. Blood pressure can become too

low, leading to dizziness or fainting. Blood sugar levels may drop. Women also may note menstrual changes. Less often, joint pain, muscle aches, fever, mental changes, or elevations of calcium may be noted.

- *Stress* - is a fact of nature in which forces from the inside or outside world affect the individual. The individual responds to stress in ways that affect the individual, as well as their environment. Due to the overabundance of stress in our modern lives, we usually think of stress as a negative experience, but from a biological point of view, stress can be a neutral, negative, or positive experience.
- *Stroke* - also known as a cerebrovascular accident or CVA, occurs when part of the brain loses its blood supply and the part of the body that the blood-deprived brain cells control stops working. This loss of blood supply can be ischemic because of lack of blood flow, or hemorrhagic because of bleeding into brain tissue. A stroke is a medical emergency because strokes can lead to death or permanent disability, plus there are now opportunities to treat ischemic strokes but that treatment needs to be started in the first few hours after the signs of a stroke begin
- *Hepatitis C infection* - is an infection of the liver caused by the hepatitis C virus (HCV). It is difficult for the human immune system to eliminate hepatitis C from the body, and infection with help C usually becomes chronic
- *Microcephaly* - is a medical condition in which the circumference of the head is smaller than normal because the brain has not developed properly or has stopped growing.
- *Low blood pressure* - also called hypotension, is blood pressure low enough that the flow of blood to the organs of the body is inadequate and symptoms and/or signs of low blood flow develop shock.
- *An aortic aneurysm* - involves the aorta, the major artery that leaves the heart to supply blood to the body. An aortic aneurysm is dilation or bulging of the aorta.
- *Others* – child abuse, nose bleeding, brain damage, alcoholism, cancer, Hepatitis B, chronic pain, suicide, antisocial personality disorder, learning disability, sexual problems, anxiety, hypersomnia, hyper sensitivity, domestic violence,

Behavior change communication

Behavior change communication (BCC) is an interactive process of any intervention with individuals, communities to develop communication strategies to promote positive behaviors which are appropriate to their settings.

****Group work*******

What are the importance of behavior change communication?

Process

- Situation analysis
- Audience analysis
- Content analysis
- Problem statement

Message development process

- Existing behaviors
- Desired behaviors
- Benefits of desired behavious
- Difficulties of desired behavious

- Alternative behavious

Stages of change:

- Awareness
- Accept responsibility
- Perception of options
- Evaluate options
- Making choices
- Experiment behavious
- Habitual behavior

Audience segmentation:

- People mostly affected
- People directly influencing them
- People indirectly influencing them

Chanel mix:

- Community dialogue
- There tools
- Interpersonal
- Mass media
- Outdoor advertising

Factors affecting BCC

- Intention;
- Environmental constraints;
- Skills;
- Anticipated outcomes (or attitude);
- Norms;
- Self-standards;
- Emotion;
- Self-efficacy.

Primary Health Care

Goal: To improve the knowledge and skills on primary health care

Specific objectives:

1. To understand the meaning, purpose, origin and levels of primary health care
2. To explain the pillars and elements of primary health care
3. To understand the implementation of primary health care
4. To review the successes and failures of primary health care implementation

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Meaning – Primary Health Care is defined as an essential health care based on practical, scientifically sound, socially acceptable methods, technology made universally acceptable to individuals, families and community through their full participation and affordable cost that the community can sustain and become self-reliant

Purposes– the importance of primary health care are:

- Reach everybody – particularly those in great need
- Reach the home and family levels, and not to be limited to health facilities
- Involve a continuity relationship with persons and families
- Improving family health - with particular focus on mothers and children
- Increase coverage and accessibility of essential health services
- Improve the quality of services
- Pursue an integrated inter-sectorial and multi-disciplinary approaches
- Increase community participation in the planning, delivery, and monitoring of health services

Origin – The concept of PHC started as follows:

- The Declaration of Alma-Ata was adopted in September 1978 at the International Conference on PHC at Almaty (formerly Alma-Ata) in Kazakhstan (formerly Kazakh Soviet Socialist Republic)
- This was the first international declaration underlining the importance of primary health care
- Goals and targets were set for achieving ''Health for All'' by the Year 2000. Some of these goals were that:
  - ✓ At least 5% of gross national product should be spent on health;
  - ✓ At least 90% of children should have a weight for age that corresponds to the reference values;
  - ✓ Safe water should be available in the home or within 15 minutes' walking distance, and adequate sanitary facilities should be available in the home or immediate vicinity;
  - ✓ People should have access to trained personnel for attending pregnancy and childbirth; and
  - ✓ Child care should be available up to at least one year of age.
- It expressed the need for urgent action by all governments, all health and development workers, and the world community to protect and promote the health of all people
- The PHC approach has since then been accepted by member countries of the WHO as the key to achieving the goal of "Health for All".

Principles – A principle is a rule that has major influence on the way in which something is done. Therefore, the principles of PHC are rules or guidelines that govern the implementation of PHC's activities. The five basic principles are:

- *Equitable distribution*
  - ✓ Equity is the fair distribution of available resources to all individuals and families so that they can meet their fundamental and basic needs.
  - ✓ Services should be physically, socially, and financially accessible to everyone
  - ✓ People with similar needs should have equal access to similar health services
  - ✓ Equal access, resource distribution and coverage of PHC services should be greatest in those areas with the greatest need

- *Manpower development*
  - ✓ PHC aims at mobilizing the human potential of the entire community by making use of available resources
  - ✓ This principle facilitate the identification and development of necessary personnel as well as training new categories of health workers to serve the community

- *Community participation*
  - ✓ This is the process by which individuals, families and communities take responsibility in promoting their own health and welfare
  - ✓ Full participation is important in decision making and taking care of own health
  - ✓ Community members and health providers need to work together in partnership to seek solutions to problems facing the communities
  - ✓ Communities should participate in: Creating and preserving healthy environment; Maintaining preventive and promotive health; Sharing information about own needs with higher authorities; Implementing health care priorities and managing health facilities

- *Appropriate technology*
  - ✓ This is a kind of technology that is scientifically or technically sound and adoptable to local needs, and the community can afford to maintain it for their self-reliance
  - ✓ Caregivers should be trained to deliver services using the most appropriate and cost-effective methods and equipment for their level of care
  - ✓ Full participation is important in decision making and taking care of own health

- *Multi-sector of Inter-sector approach*
  - ✓ Health sector cannot achieve much in isolation
  - ✓ PHC requires coordinated effort with other related sectors whose activities impacts on health e.g., agriculture, water and sanitation, transportation & communication, education etc.
  - ✓ The commitment of all sectors may increase if the purpose for joint actions and roles of each sector is clearly defined

Levels – A level is a stage or position of something. Levels of PHC are the units of care or implementation of activities. The levels of health care are:

- Individual Based Health Care
- Family Based Health Care
- Community Based health care
- Facility Based Health Care(Dispensary, Health center, District hospital, Provincial hospital)

Pillars – A pillar a column or support. Pillars of PHC are those support systems that make it achieve goal of improved health by creating partnership between consumers of health services with health care providers. The Seven pillars are:

- Health systems
- Priority
- Science
- Culture
- Equity
- Participation
- Sustainability

Elements – Are fundamentals or essentials of PHC. In In the Alma Ata conference of 1978, eight essential elements of PHC were identified. However, individual countries were given the liberty to add any other elements they felt were relevant to their own country. The PHC elements listed at the Alma-Ata declaration were as follows:

1. Health education
2. Local disease control
3. Expanded programme of immunization
4. Maternal and child health care and family planning
5. Essential drug supply
6. Nutrition and adequate food supply
7. Treatment and prevention of common diseases and injuries
8. Safe water supply and good sanitation

Kenya government has added more elements. These are:

1. Mental Health
2. Dental Health
3. Community Based Rehabilitation
4. HIV/AIDS Prevention
5. Health information systems

Strategies – Are approaches or methods of implementing an activity. There are three main PHC strategies:

- *Community Based Health Care*
  - ✓ These are health care activities initiated and implemented by community own resource persons
  - ✓ It is for people of all ages who need health care assistance at home. Community care services include home support, home nursing, physiotherapy and other rehabilitation services.
  - ✓ For example: A CHW visits an elderly person at home to help with medication, or a CHW support worker helps a child in a wheel chair with personal care, such as bathing.

- *BAMAKO Initiative*
  - ✓ This initiative was a formal statement adopted by African health ministers in 1987 at Bamako in Mali.
  - ✓ It was meant to implement strategies designed to increase the availability of essential drugs and other health care services for Sub-Saharan Africans
  - ✓ This was an initiative of drug revolving funds as income generating activity to sustain PHC activities.

- *Community strategy*
  - ✓ This is the concept of taking health services close to the people in the village
  - ✓ It is aimed at taking the Kenya Essential Package for Health to the community. It is a strategy for the delivery of level one service
  - ✓ The overall aim is to involve the communities in addressing the downward trend of deteriorating health status. It is a mechanism through which households and communities take an active role in health and health-related development issues.

Roles – Roles are responsibilities assigned to an individual or a group with an aim of achieving a desired objective. The following are key roles of Health Records & Information Officers and Community Health Workers in PHC implementation:

- *Health Records and Information Officer*
  - ✓ Designing, developing and distribution of PHC data collection tools
  - ✓ Data collection, analysis and dissemination
  - ✓ Training of CHW on data collection and management
  - ✓ Advising the PHC team on targets and achievements of PHC
  - ✓ Monitoring and evaluation of PHC implementation

- *Community Health Workers*
  - ✓ Identify and refer patients
  - ✓ Treatment of minor illnesses
  - ✓ Cary out health education
  - ✓ Defaulter tracing
  - ✓ Support home based care services
  - ✓ Home follow up

Success & failures – Are attainment of achievements or inability to attain any intended achievement in any endeavor. The following are reasons why PHC did not achieve its intentions in most of the cross-cutting countries:

- Many ordinary people felt PHC was a cheap form of healthcare and, if they were able to, they bypassed this level to attend secondary and tertiary centers because of a lack of staff and essential medicines at the PHC level.

- Civil war, natural disasters and, more recently, HIV affected the ability of PHC to maintain comprehensive services, especially in many sub-Saharan countries.

- Political commitment was not sustained after the initial euphoria of Alma-Ata. In many cases PHC became a jargon term used as a slogan, and little else. The rhetoric was not backed with the necessary reforms. Agencies were content if countries adopted PHC as a policy, and did not assess actual practice. Politicians saw PHC as a way to reduce expenditure in health and lacked the political will to ensure that services were equitably shared and distributed. Most healthcare resources continue to be directed to the large urban-based hospitals.

- Issues of governance and corruption in the use of resources resulted in donors becoming very wary of funding comprehensive, broad-based programs. Vertical, definable, time-limited programs that could be changed every few years suited both donor agencies and governments.

****** **Field Visit to a nearby PHC site** *****
Visit a nearby PHC site and identify:

1. **What is the name of the site?**
2. **When did the site start?**
3. **How many CHWs are active?**
4. **What are the PCH services offered?**
5. **What are the 10 factors that are making PHC successful in the site?**
6. **What are the 10 factors that are making PHC fail in the site?**
7. **What are the plans to revitalize PHC activities in the site?**

Maternal and child health

Goal: Understand the concept of facility and community based maternal and child health care

Specific objectives:

- Outline the goal and focus of MCH and family planning
- Describe MCH and family planning services
- Describe community base approach to MCT and FP

Introduction

- Maternal health is the health of women during pregnancy, childbirth, and the postpartum period.
- It encompasses the health care dimensions of family planning, preconception, prenatal, and postnatal care in order to reduce maternal morbidity and mortality.
- Preconception care can include:
    - Education,
    - Health promotion,
    - Screening
    - and other interventions among women of reproductive age to reduce risk factors that might affect future pregnancies.

- Postnatal care issues include:
    - Recovery from childbirth,
    - Concerns about newborn care,
    - Nutrition,
    - Breastfeeding,
    - Family planning

- The goal of prenatal care is to:
    - Detect any potential complications of pregnancy early,
    - Prevent them if possible,
    - Direct the woman to appropriate specialist medical services as appropriate.

- Primary areas of focus in MCH include:
    - Increasing healthy birth outcomes;
    - Promoting and assuring comprehensive primary care for children, including children with special health care needs;
    - Promoting healthy lifestyles among school-age youth, ages 6-21, including children with special health care needs;
    - Assuring access to safe, healthy child care up to age six, including children with special health care needs;
    - Preconception, pregnancy testing and planning.

- In Africa, one out of 210 mothers dies during pregnancy or delivery.
- Most maternal deaths are due to causes directly related to:
    - Pregnancy
    - Childbirth,
    - Unsafe abortion
    - Obstetric complications such as:

        - Severe bleeding
        - Infection
        - Hypertensive disorders
        - Obstructed labor
    - Other indirect cause like:
        - Relatively low rate of institutional deliveries,
        - Due to transport problems
        - Lack of infrastructure,
        - Due to cultural prejudices and resistance against giving birth outside the family circle

- The fifth Millennium Development Goal (MDG) aims to reduce the maternal mortality ratio by 75% between 1990 and 2015.
- In Kenya, maternal mortality remains high at 488 maternal deaths per 100,000 live births.
- The MCH/FP clinic is founded on quality and afford-able services that advocate for safe motherhood, proper child development, and a healthy family.

MCH Services

- The services offered in MCH clinic are as follows:
    - Focused antenatal care.
    - Targeted post natal care.
    - Providing all family planning methods: short term, long term, permanent i.e. vasectomy & tubal ligation.
    - Screening of Breast and Cervical cancer in all women of reproductive ages.
    - Giving immunization i.e. BCG, Pentavalent, Polio, Measles.
    - Growth monitoring for children under the age of five.
    - Counseling and testing of antenatal mothers and partners on PMTCT - eMTC
    - Create awareness through Micro teaching on various topics i.e. Nutrition, Breast feeding, Weaning etc.

Immunization

- The basic principle of immunization is to administer into a healthy person a vaccine that will prevent that person from getting a certain disease.
- Vaccines may be of:
    - Live attenuated: (e.g. Rubella, OPV, Measles and BCG),
    - Inactivated or killed (e.g. Hib, IPV)
    - Micro-organisms
    - Detoxified toxins (e.g. Tetanus)
- Generally, several vaccines can be given at the same time. This is important since you do not know when you will see the child again.
- BCG, OPV, DPT-HeB-Hib and Measles vaccines can be given simultaneously if the child is of the appropriate age and has not received the early immunizations.
- A critically ill child needing hospital admission must be given the appropriate vaccines upon recovery
- Remember:
    - That a slight fever and/or other minor illness should not prevent you from immunizing a child
    - To inform mothers/child-caretakers about possible side effects of each of the given vaccines

- To record all vaccinations on tally sheets and on the Child-Health immunization cards and instruct the mothers always to bring the cards along with them when taking children to a health facility
- To instruct mothers to return the child for the next immunization date as indicated on the card
- That vaccines are easily destroyed by heat and rendered ineffective
- To handle the disposal of used sharp syringes appropriately
- To ensure appropriate cold storage of the vaccines and follow the recommended cold- chain instructions for each of the vaccines carefully.

Facts about the common vaccine-preventable diseases
Whooping cough (Pertussis):

- A communicable disease.
- Spread by droplets.
- Symptoms include:
    - Severe cough followed by a whoop and vomiting, Leads to malnutrition, Can cause death, Severe in under 1 year old.

Diphtheria:

- An infectious disease.
- Spread by droplets.
- Symptoms include: Difficulty in breathing, Swallowing, Enlarged neck.
- Very severe when it occurs.

Tetanus: [see tetanus]:

- A clinical syndrome involving primarily the central nervous system and resulting from the tetanus toxins.
- Enters through open wounds, cuts and umbilical stump.
- Symptoms include: Stiffness, Locked jaw, Inability to suckle and muscle spasms.
- Has a very high mortality (>50%).
- Immunizing pregnant mothers ensures protection of her new born baby.

Measles (Rubeola):

- Highly infectious killer disease.
- Symptoms include: Rash, Fever, Cough, Red eyes;
- Is associated with: blindness, malnutrition, deafness, pneumonia and death.

Poliomyelitis (Infantile Paralysis):

- An acute communicable disease.
- Spread by droplets and oro-faecal contamination.
- Symptoms include:
    - Pain and flaccid paralysis in limbs, fever, vomiting; can lead to permanent deformity, and can cause death.

Tuberculosis:

- A communicable disease.
- Spread by droplet.

- Symptoms include: fever, wasting, deep chesty cough, Night sweats, May have lymphadenopathy.
- Leads to lowered resistance to other diseases and may be fatal.

Hepatitis B:

- Is a highly infectious disease.
- Transmitted mainly by:
  - Parenteral route,
  - From person to person
  - By close contact
  - Through exchange of body fluids such as:
    - Saliva, Secretions from open wounds, Blood, Vaginal secretions and semen.
- Transmission between children is common, since they are often more infectious than adults.
- Transmission from carrier mothers can occur in up to 80% of babies during the perinatal period,
- Infection may be transmitted either:
  - Vertically (transplacentally from mother to unborn baby)
  - Horizontally by close contact.

Childhood immunization schedule in Kenya

Vaccine	Age	Remarks
BCG POLIO (OPV O)	At birth Birth Dose	Or at first contact with child
DPT_1-HeB_1-Hib_1 DOSE POLIO (OPV 1)	6 weeks (1 ½ months)	Or at first contact with child after that age
DPT_2-HeB_2-Hib_2 DOSE POLIO (OPV 2)	10 weeks (2 ½ months)	4 weeks after DPT 1 and OPV 1 can also be given anytime after this period, when in contact with the child.
DPT_3-HeB_3-Hib_3 DOSE POLIO (OPV 3)	14 weeks (3 ½ months)	4 weeks after DPT 2 and OPV 2 can also be given any time after this period, when in contact with the child.
Measles	9 months	May be given between 6 and 9 months if child is admitted to hospital for any other illness. Repeat at 9 months as per KEPI schedule.

Immunization schedule for pregnant mothers (TT2+)

Tetanus Toxoid (TT)	Pregnant women	1st dose with first pregnancy or subsequent pregnancy
		2nd dose - 4 wks after first dose
		3rd dose - 6 months after 2nd dose
		4th dose - at least 1 year after the third dose

5th dose - at least 1 year after the fourth dose

Immunizing a pregnant mother ensures protection of her newborn baby.

Vitamin A supplementation schedule

Dosage (in U)	When to give
<6 months - 50,000	First dose at 9 months then every 6 months (twice per year) up to the age of 60 months.
6-12 months- 100,000 iu	
>12 months 200,000 iu	

VACCINE DOSAGE AND ROUTE OF ADMINISTRATION

VACCINE DOSE (*=Live vaccines)	ROUTE OF ADMINISTRATION
BCG* Child under 1 year, 0.05 ml. Child over 1 year, full dose, 0.1 ml	Intra-dermally into upper outer part of left forearm, at the junction of the upper and middle thirds. If given correctly a small wheal appears at the site of the injection. Inform the mother that a small sore will appear in 2-6 weeks. Let this heal by itself. It will leave a small scar. If no reaction develops, the vaccination should be repeated after 3 months.
POLIO (OPV)* 2 drops by mouth. Follow manufacturer's instructions on dosage	Read the instructions on the bottle. Give vaccine by mouth. Use dropper provided. If child spits or vomits repeat the dose.
Pentavalent vaccine consisting **of DPT-HeB-Hib** 0.5 ml	Intramuscularly in the upper outer part of the thigh.
HB 0.5 ml - child 1.0 ml - adult	Intramuscularly in the upper outer part of the thigh for child and deltoid (left) for adult
Measles* 0.5 ml	Subcutaneously or intramuscularly in upper outer part of the arm (deltoid muscle)
Tetanus Toxoid (TT) 0.5 ml.	Intramuscularly in the outer part of the upper-arm (deltoid muscle)

Notes: Hands should be washed before and after handling vaccines. All the vaccines and diluents must be kept cold. DPT, HB, and TT vaccines are damaged if kept below 0°C and therefore should never be frozen. Always check Vaccine Vial Monitor (VVM).

Contra-indications. A definite severe reaction to a preceding vaccine dose is a contraindication to further doses.
Live vaccines should not be given to individuals with impaired immune response e.g. leukaemia, HIV/AIDS.

The intramuscular route should not be used in patients with bleeding disorders such as haemophilia or thrombocytopaenia.

HIV/AIDS INFECTION AND IMMUNIZATION
Asymptomatic children infected with HIV should receive all standard Kenya Expanded Programme on Immunization (KEPI) vaccines. Non-immunized children with symptomatic HIV infection should receive all standard vaccines except BCG.

Side-effects and adverse reactions to vaccinations
Range from mild to severe for various vaccines.

BCG Vaccine
These include injection abscess, regional or wide-spread lymphadenitis, osteomyelitis and disseminated BCG infection. These should be treated with anti-tuberculosis drugs **[see 12.8. TB]**

Oral Polio Vaccine
Adverse reactions rarely occur.

Measles Vaccine
Adverse reactions include fever, mild rash and rarely convulsions, encephalitis and sub-acute schlerosing pan-encephalitis (SSPE).

DPT (Diphtheria, Pertussis, Tetanus)
Most adverse reactions are attributed to the pertussis component. Minor reactions include, pain at the site of injection and fever. Major reactions are persistent crying, high pitched cry, excessive somnolescence, convulsions, encephalopathy and coma.

Recombinant DNA Hepatitis B Vaccine
Side effects include pain, fever and swelling at the site of injection.

Family Planning and Contraceptive methods, types and Options in Kenya

- The choice of one method of contraception against the other depend on:
  - individual medical condition
  - taste of an individual or partner,
- The most common method of family planning is hormonal method except condom followed by mechanical methods. The most reliable method is surgical method and the most unreliable method is natural methods.
- Type of family planning and contraceptives can largely be divided into four categories

- Natural
- Barrier
- Hormonal
- Surgical

NATURAL METHODS OF FAMILY PLANNING AND CONTRACEPTIVES METHODS IN KENYA

- It is called natural because it does not involve use of drug or device
- The following are the available method or types of natural family planning methods
- The following make up natural family planning methods:
  - **Withdrawal Method (coitus interrupters)**
    - Warning this method can only be used in highly motivated couples.
    - It is useful as a temporary method for example supplies for of regular method are not available or the couples are together for only a short time.
  - **Safe period (rhythm method)**
    - This one means that couples avoid sex the time the woman eggs (ovum) are ready for fertilization.
    - This is around mid-month of a woman's circle.
    - This means that there should no sex for the couples who are using this method between day 8(eight) and day 20 (twenty) of a woman's circle i.e. a total of 12 days without sex in a month.
    - Have you ever seen a cow on heat? Unfortunately this is the period at which women folks are on heat so this method like the withdrawal method require highly motivated couples
  - **Breastfeeding amenorrhea method**
    - Risk of pregnancy in exclusively breastfeeding couples in the first 6 (six) months so long as the woman has not experienced menses is 2 % (percent)
    - It is a reliable method so long as the above conditions are intact.
  - **Cervical mucus membrane**
    - This is a slightly technical method and couples are advised against using this if they don't understand the physiology of ovulation.
  - **Basal body temperature (BBT) method**
    - Like cervical mucus membrane this is also a slightly cumbersome method because it involve daily recording of a chart.
    - Check the temperature five minutes before waking up and enter in ascertain chart after woman's eggs are released (ovulation) temperature rises by (0.2 – 0.6) C intercourse should be avoided when the temperature is low and resumed when the temperature has risen up next period.
    - like you might have noticed this method is very unreliable form of family planning, what if one gets sick an the temperature rises, what if one fails to record the temperature.
    - This method can be easily manipulated by women folk to deny men folks there conjugal rights by just manipulating the temperatures.

Note: Natural family planning requires high motivation and so makes them not particularly reliable method of family planning.

BARRIER METHODS OF CONTRACEPTION AND FAMILY PLANNING IN KENYA

- These method or type of family planning include:
  - Male and Female Condoms
  - Spermicides
  - Diaphragms
  - Cervical Caps
- Whereas condoms, diaphragms, and cervical caps are mechanical barriers, spermicides are chemicals that interfere with the movement of the sperm and its ability to fertilise the egg.
- Currently in Kenya, the use of diaphragms, cervical caps, and spermicides is negligible.
- In addition, scientific evidence has shown that repeated and high-dose use of the spermicide might cause vaginal and cervical irritation or abrasions, which could increase the risk of infection with HIV.

HORMONAL METHODS OF FAMILY PLANNING AND CONTRACEPTIVE METHODS IN KENYA

- This is the most commonly used method or type of family planning and contraceptive after condom
- They include hormone taken orally, by injection or implanted to prevent development and maturation of woman's eggs (ovum's)
- Common hormonal FP and contraceptives methods are:
  - Emergency contraception (e-pills)
  - Ordinary pills
  - Injectable contraceptives
  - Contraceptive Implants (Jadelle, Implanon, Zarin)
  - Hormone-releasing intrauterine systems
    - This method involves use of devices which may sometime have some form of drug (hormone) attached to it ,
    - Their mode of function is by placing a barrier so that sperm does not get into the fallopian tube to fertilize the woman's egg.
    - The following are the available IUCD type of family planning and contraceptives methods in Kenya
      - Copper-based IUCDs
      - Hormone-releasing devices

VOLUNTARY SURGICAL CONTRACEPTION

- Apart from use of condom there is another area in which men can contribute to family planning directly.
- The following are the available surgical family planning methods:
  - Tubal legation
  - Vasectomy

Community-Based Maternal, Newborn and Child Health Services

- Aims to reduce the number of deaths among mothers, newborns, and young children by improving the health services available in four rural districts

- Provides women with greater access to good-quality, reliable health services.
- It helps to ensure that district health facilities offer essential health services for women and children;
- It provides training for health care workers in emergency care for pregnant women, new mothers, and newborn babies;
- It strengthens the management and supervision practices in health care facilities.
- It improves the ability of men and women in the communities to recognize, prevent, and respond to maternal and child health issues.
- The community MCH services include:
    - Organizing community events and mass media campaigns,
    - Training community theatre groups,
    - Selecting peer youth educators and male champions to raise awareness about potential health issues;
    - Conducting home visits to provide pre-natal and post-natal care and care for newborns and young children;
    - Equipping community health workers to provide services in remote areas and to refer patients to health centres where necessary.
- It emphases the role that men can play in helping to improve the health of women and children.

Growth Monitoring

- This is the monitoring of the growth of a child through nutrition.
- Weight of the child is monitored using the CWC card
- The weight of the child is monitored for 5 years

Nutrition and Health

Goal: Increase understanding on nutrition in the context of community health

Specific objectives:

- Describe nutrition and dietary requirements in our lives
- Describe the implications of malnutrition in health and development

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Nutrition - Nutrition is the intake of food, considered in relation to the body's dietary needs. Good nutrition is an adequate, well balanced diet combined with regular physical activity. It is the cornerstone of good health. Poor nutrition can lead to reduced immunity, increased susceptibility to disease, impaired physical and mental development, and reduced productivity. Eating well is important for everyone's health and wellbeing and is vital for preventing chronic disease in later life.

Community nutrition – is the nutrition in the context of community. In order for nutrition to have impact on human health and well-being, it must be applied to human populations through community and population based intervention and education programs.

Malnutrition - Malnutrition is the condition that develops when the body does not get the right amount of the vitamins, minerals, and other nutrients it needs to maintain healthy tissues and organ function. It is a cause and consequence of ill health across many groups; however, older people tend to be more vulnerable to this condition, which can severely impact on health outcomes and quality of life.

Basic nutrition - Basic nutrition is the food intake, dietary nutrients required during our phases of life

- *Food intake* – An average person takes at least 3 meals in a day (breakfast, lunch and supper). People tend to have supplementary feeds in between the main meals. Children are required to be fed every time according to their demand. Food intake is influenced by income, appetite, exposure to food and social status.
- *Dietary nutrients* – Our body requires essential dietary nutrients for its normal function. The nutrient requirements are: Carbohydrates, Proteins, Water, Vitamins, and Minerals. Essential nutrient are substance that must be obtained from the diet because the body cannot make it in sufficient quantity to meet its needs. In establishing dietary requirements the goal is to determine what amount is high enough, if ingested routinely, to prevent impairment of health even if intake becomes inadequate for a short period
- *Nutrition during phases of life* – This is the nutrition required during: Pregnancy and early childhood; Childhood and adolescent; School going age; Adults; and Older age.
    - *Pregnancy and early childhood* - Starting early with breastfeeding and providing good nutrition for children gives the best start to a healthy life. Good nutrition in pregnancy is essential for keeping a developing baby and mother healthy. Breastfeeding is the natural way to feed babies. Breastfeeding is extremely important for the health outcomes of infants and their mother. Not breastfeeding is associated with adverse health outcomes for both mothers and infants, ranging from infectious illness to chronic diseases later in life. Solids should be introduced at around six months of age when the infant is developmentally ready. For young children, a healthy diet is required for optimal growth and development. It can also help to improve concentration, fight illnesses, and prevent long term health complications.

    - *Children and Adolescents* - Children and adolescents need sufficient foods to grow and develop normally. Growth should be checked regularly for young children. Physical activity is important for children and adolescents. Many infants suffer and die because of ignorance, meanness of parents and inability to understand the infant communication language

- *School aged children* - Childhood is an important period when children develop the knowledge, skills and behaviors for lifelong health and wellbeing. We know that education and health are inseparable and that a healthy child learns well.

- *Adults* - Healthy eating during adulthood is vital for effective functioning of body systems and immunity, maintaining a healthy weight, and preventing and managing chronic conditions such as diabetes and heart disease.

- *Older people* - Good nutrition is important for older people to help promote independence and to reduce illness and premature death. Maintaining a healthy lifestyle, including good nutrition, plays an important role in ensuring older people can live well at home for longer.

Components of community nutrition

Community nutrition entails the following components:

- *Surveillance of food chain* – This is assessing food for possible infestation with germs, toxic. It's also called food contamination monitoring
- *Nutritional epidemiology* - A medical research that studies the relationship between nutrition and health. It is the on effects of dietary intake and nutritional status on health of individual or community. Multiple nutrients work together in the body, with differing impacts on different systems and under differing environmental conditions.
- *Clinical nutrition and diabetes* - The number of patients with diabetes is growing rapidly. Involves assessing the effectiveness of *clinical nutrition* in diabetes care and outcomes
- *Nutrition education* **-** Is any combination of educational strategies, accompanied by environmental supports, designed to facilitate voluntary adoption of food choices and nutrition-related behaviors conducive to health and well-being. Nutrition education is delivered through multiple venues and involves activities at the individual, community, and policy levels
- *Food and nutrition policy* – It make healthy food more available; Disclose the calorie content of restaurant foods; Teach people how to make healthy eating easier. There are 3 existing nutrition policies and programs, like: Nutrition Facts labels on packaged foods; Nutrition standards for school lunches; Regulation of food additives.
- *Food industry and standard agency* - is a complex, global collective of diverse businesses that supply much of the food and food energy consumed by the world population e.g. UNILIVER. The Food Standards Agency is concerned with the whole food industry – from farming, food production and distribution, to retail and catering. It addresses food safety issues at every stage of the chain, providing information and guidance on best practice and legal requirements.
- *Mass catering* - Food and drink supplied by a caterer to large numbers of customers at the same time

Malnutrition

- Individuals are malnourished, or suffer from under nutrition if their diet does not provide them with adequate calories and protein for maintenance and growth, or they cannot fully utilize the food they eat due to illness.
- Growth failure malnutrition, as the name suggests, is the failure of an individual to grow as expected in stature or weight, according to his/her age and gender. Growth failure malnutrition can take three serious types:
    - Acute malnutrition or wasting arises out of sudden, drastic weight loss. It leads to three clinical malnutrition types:

        - *Marasmus*: this occurs when body fat and tissues degenerate at an alarming rate to compensate for the lack of nutrients. As a result, the body's internal processes begin to slow down alarmingly fast as does the activity of the immune system.
        - *Kwashiorkor*: this is characterized by bilateral pitting oedema (fluid retention) in the legs and feet. As a result, the under-nourished child may actually look plump.
        - *Marasmic-kwashiorkor*: This is characterized by both severe wasting and oedema.

    - Chronic malnutrition or stunting happens over a long period of time and has more long-lasting consequences. It begins before birth due to poor maternal health and leads to stunted growth in an otherwise normally proportioned child. Poor breast feeding, infections and lack of availability of proper nutrients are the main causes behind it. Stunting is dangerous because it becomes irreversible after an age. Therefore, it becomes important to nip it in the bud by providing proper medical treatment to pregnant women and young girls.

    - Micronutrient malnutrition implies a moderate to severe lack of Vitamins A, B, C and D, Calcium, Folate, Iodine, Iron, Zinc and Selenium. These vitamins and minerals are of utmost importance in various body processes and their deficiency can make an otherwise healthy person malnourished:
        - Iron deficiency causes anaemia, poor brain development and cardiac functioning.
        - Iodine deficiency leads to impaired thyroid functioning and mental retardation.
        - Vitamin D deficiency causes rickets and other bone development related disorders.
        - Selenium deficiency leads to poor cardiac function, weak immunity and osteoarthritis.
        - Vitamin A deficiency is a cause of poor vision, bone development and immunity.
        - Vitamin B12 deficiency leads to nerve degeneration and poor RBC formation
        - Folate or vitamin B9 deficiency causes slow growth and anemia
        - Zinc deficiency can cause poor immunity, sensory perception and anemia.

- Signs and symptoms of malnutrition- A symptom is something the patient feels and reports, while a sign is something other people, such as the doctor detect. For example, pain may be a symptom while a rash may be a sign. The general signs and symptoms include:
    - Loss of fat (adipose tissue)
    - Breathing difficulties, a higher risk of respiratory failure
    - Depression
    - Higher risk of complications after surgery and lower healing hence longer recovery
    - Higher risk of hypothermia - abnormally feeling cold and low body temperature
    - The total number of some types of white blood cells falls; consequently, the immune system is weakened, increasing the risk of infections.
    - Lower sex drive and problems with fertility
    - Reduced muscle and tissue mass
    - Tiredness, fatigue, apathy and irritability
    - In more severe cases:
        - Skin may become thin, dry, inelastic, pale, and cold

- Eventually, as fat in the face is lost, the cheeks look hollow and the eyes sunken
- Hair becomes dry and sparse, falling out easily
- Sometimes, severe malnutrition may lead to unresponsiveness (stupor)
- If calorie deficiency continues for long enough, there may be heart, liver and respiratory failure
- Total starvation is said to be fatal within 8 to 12 weeks

- Causes of malnutrition are:
  - *Poor diet* - if a person does not eat enough food, or if what they eat does not provide them with the nutrients they require for good health, they suffer from malnutrition. Poor diet may be caused by one of several different factors. If the patient develops dysphagia (swallowing difficulties) because of an illness, or when recovering from an illness, they may not be able to consume enough of the right nutrients.
  - *Mental health problems* - some patients with mental health conditions, such as depression, may develop eating habits which lead to malnutrition. Patients with anorexia nervosa or bulimia may develop malnutrition because they are ingesting too little food.
  - *Mobility problems* - people with mobility problems may suffer from malnutrition simply because they either cannot get out enough to buy foods, or find preparing them too arduous.
  - *Digestive disorders and stomach conditions* - some people may eat properly, but their bodies cannot absorb the nutrients they need for good health.
  - *Alcoholism* - this is a chronic (long-term) disease. Individuals who suffer from alcoholism can develop gastritis, or pancreas damage. These problems also seriously undermine the body's ability to digest food, absorb certain vitamins, and produce hormones which regulate metabolism. Alcohol contains calories, reducing the patient's feeling of hunger, so he/she consequently may not eat enough proper food to supply the body with essential nutrients.
  - Food shortage, price and distribution
  - Lack of breastfeeding

Impact of Malnutrition on Health and Development

- Malnutrition and starvation have devastating impact on children, adults and especially on pregnant women. They also have severe and far-reaching socio-economic impacts.
  - *Effects on children* - Mental retardation, stunted growth, poor immune system, micronutrient deficiency, GI tract infections, anemia and inevitably – death.
  - *Effects on pregnant women* - Besides the health of the child being poor, a pregnant woman might have difficult labour, postpartum hemorrhage and anemia.
  - *General effects* - Weak immunity, inactivity of muscles, apathy, depression, and kidney function impairment.

- Malnourished children experience developmental delays, weight-loss and illness as a result of inadequate intake of protein, calories and other nutrients. Because orphaned and institutionalized children may experience one or several macronutrient and micronutrient deficiencies, they are at risk for a variety of short-term and long-term complications:

- Short-Term Implications - Because so much development occurs in the first few years of life, nutrient deficiencies can have major short-term implications in young children.
  - **Immune Implications -** Malnourishment can greatly compromise a child's immune system, making them more susceptible to infectious diseases. Particularly in institutions where there are poor sanitary practices, children are vulnerable to infections from other children or caregivers.

  - **Growth Implications -** A child may contract an infection due to poor nutritional status. In turn, a gastrointestinal infection places the child at even greater risk for nutrient deficiencies because nutrients are unable to be absorbed properly. Consequently, nutrient deficiency combined with infection can cause growth retardation.

  - Additionally, a deficiency in one nutrient may lead to a deficiency in another nutrient. For example, deficiencies in iron, magnesium and zinc can cause anorexia and thereby result in reduced intake of other important nutrients such as protein. Low lipid intake can also affect the absorption of important fat-soluble vitamins such as vitamins A and Zinc and protein deficiencies can retard bone growth and development, putting a child at risk for long-term complications.

- Long-Term Implications - The short-term implications of malnutrition eventually give way to long-term complications, such as growth and cognitive delays:
  - Growth Implications - Malnutrition not only impacts growth in the short term, but can also limit total bone growth. Additionally, children classified as low height-for-age (stunted) may never be able to regain lost growth potential if they continue to live in a nutritionally deprived situation.
  - **Cognitive Implications -** Malnutrition negatively effects brain development causing delays in motor and cognitive development, such as:
    - Attention deficit disorder
    - Learning disabilities and impaired school performance
    - Decreased IQ scores
    - Memory deficiency
    - Reduced social skills
    - Reduced language development

********Field practice *********

Group1. Surveillance of food chain

1. What methods can you adopt in assessing food for possible infestation with germs, toxic?
2. Develop a food contamination monitoring questionnaire
3. Use the questionnaire to assess foodstuff in a local market

Group2. Nutritional epidemiology

1. Identify all the relationship between nutrition and health
2. Develop a food frequency assessment tool
3. Use the tool to assess the food frequency among pupils attending a local primary school

Group3. Clinical nutrition and diabetes

1. Develop a dummy table eliciting cases of diabetes by age and sex

2. Use the dummy table to capture the number of diabetes cases in a nearby hospital for 5 months
3. Randomize 10 cases of diabetes notes and assess presence of nutritional therapy

Group4. Nutrition education

1. Identify communication strategies that would be suitable in promoting adoption of food choice and nutritional behaviors
2. Use the communication strategy to sensitize 5 mothers with children having malnutrition related conditions in a pediatric ward in a nearby hospital
3. Use your mobile phone to take a picture of sensitization session going on

Group5. Food and nutrition policy

1. Visit a nearby hospital, identify nutrition office or officer, ask about all policy documents on nutrition
2. Make a tour in MCH clinic, medical ward and pediatric ward to identify availability of nutritional policy documents
3. Explain in plenary the content of the policy document to the class

Group6. Food industry and standard agency

1. Visit a nearby supermarket, sample 10 different foodstuff to identify the food industry that has manufactured it
2. Identify 10 foodstuff at the shelf and document all the ingredients that makes the foodstuff

- Identify if the food's safety has been certified by the Kenya Bureau of Standards.

Group7. Mass catering

1. Visit the Siaya MTC rationing store and identify different supplies that are catered
2. In each of the supply, identify the quantity and frequency of catering
3. Describe the documentation of the supplies on arrival till during usage

Planning and Evaluating Community Health Services

Goal: Increase skills and knowledge on planning and evaluating community health programs

Specific objectives:

- Describe the concept of planning in community health
- Explain goal setting and planning process of community health programs
- Describe the concept of evaluation in community health

Community Health Planning - Community health planning is a systematic process of identifying the health needs of communities and developing evidence-based recommendations and strategies to promote and improve the health of individuals and communities.

Goal - The goal of planned approach to community health services is to increase the capacity of communities to plan, implement, and evaluate comprehensive, community-based health programs targeted toward priority health problems. Planned approach to community health helps in:

- Systematic implementation of activities
- Establishing a health working teams
- Define key data to be collected and used
- Set health priorities, and design and evaluate interventions
- Identify and address priority health problems or special population to be addressed
- It can also be adapted and used by existing organizational and planning structures in the community.

A planning cycle in the development of a community health program

- A planning cycle that includes:
- A situation analysis,
- Formulation of objectives,
- Selection of strategies,
- Development of an operational plan,
- Implementation and evaluation, which lead, in turn, to a new situation analysis.

Using data to identify health problems

- On the basis of information collected during phase II, the community group needs to identify health priorities and determine which ones it wants to address first.
- As the community group reviews different types of data, it develops criteria for determining priorities within the data. For example, as members review mortality data, they might consider something a priority if:
    - It causes a large number of deaths.
    - It is a leading cause of years of potential life lost.
    - It is a greater problem in your community than in the state or nation.

- By matching the mortality data to the criteria, the group would identifying priorities for each of the other types of data

Choosing Health Priorities

Most communities do not have the resources to address all of their health problems and target groups at once. They must set priorities and plan to address some problems initially and others over time. We recommend that the community group select only one or a limited number of health problems to better focus resources in a comprehensive manner. The following are priority setting techniques:

- Problem/Preference Ranking - is a participatory technique that allows analyzing and identifying problems or preferences *stakeholder* share in order to implement adequate improvements and solutions in their community and area. First, the *stakeholders* have to decide which the most important problems they face in their community are. Afterwards, the participants have to rank these problems/preferences in regards to their importance. The result of this method provides the starting base for discussions on possible solutions to the priority problems.

- A more systematic technique is called "pairwise ranking" and uses cards to represent the different problems. The facilitator shows the "problem cards", two at a time, each time asking, "Which is the bigger problem?" As the participants make the comparisons, the results are recorded in a matrix.

The criteria attached to make up a choice are used to consider in the community action plan. Community action plans are developed on the basis of people's preferences. The problems, solutions technical inputs are arranged on the interests of the users.

Advantages

- Preference/Problem ranking method helps to quickly get a good idea of what people think are the priority problem or preferences
- Preference/Problem ranking is probably the easiest method to use and to learn more about commonly shared problems and priorities
- The criteria developed out of the ranking can be used for the community action plan of the area

Disadvantages

- It requires time and some logistic efforts to bring all relevant stakeholders together in order to rank the preferences and problems
- It may not be possible to react to all the problems discussed

As the community group works to set priorities, it must examine the intervention focus and its capacity to address one or more problems and agree on which problem or problems to address first.

Identifying which problem(s) to address first

When examining priority behavioral risk factors, you might consider the following:

- **Importance**—evidence that the behavior change will make a difference…..eg. How widespread is the behavior? - How serious are the health consequences? How close is the connection between the behavior and the health problem?
- **Changeability**—evidence that the behavior is amenable to change ……e.g. Is the behavior still in developmental stages? Is the behavior only superficially tied to lifestyle? Has the behavior been successfully changed in other programs? Does the literature suggest that the behavior can be changed?

Many community groups find it helpful to use the nominal group process or other techniques to identify the top three to five health problems. These three to five problems are then analyzed further to determine which to address first. Some issues to consider when ranking problems include:

- Legal and economic factors,

- Political viability of the intervention,
- Possibility of continued funding,
- Probability of quick success,
- Ability to build on community strengths,
- Level of public concern.

Goal setting

Goal setting involves establishing specific, measurable, achievable, realistic and time-targeted (S.M.A.R.T) goals. Setting goals helps people work towards their own objectives. Goal setting features as a major component of personal development literature.

- Characteristics of a good goal:
  - **S**pecific - When setting goals, they should answer the highly specific questions of who, what, where, when, and why.
  - **M**easurable - In order for us to track our progress, goals should be quantifiable.
  - **A**ttainable - It is important to evaluate your situation honestly and recognize which goals are realistic, and which are a little far-fetched.
  - **R**elevant - Is this goal relevant to your life? Does this match my needs?
  - **T**ime-related - Setting a "due date" to meet goals not only keeps you on track, but it prevents pesky daily roadblocks from getting in the way.
- Facts about goal setting include:
  - Specific, realistic goals work best
  - It takes time for a change to become an established habit
  - Repeating a goal makes it stick
  - Pleasing other people doesn't work
  - Roadblocks don't mean failure
- How to set goals:
  - *Think about the "big picture***.**" - Ask yourself some important questions about what you want for your life.
  - *Break the "big picture" down into smaller and more specific goals* - Consider areas of your life that you either want to change or that you feel you would like to develop with time. Begin to ask yourself specific questions about what you'd like to achieve in each area and how you would like to approach it within a five year timeframe.
  - *Make each goal a positive statement* **-** Using positive statements is a direct way to affirm your commitment to completing your goals.
  - *Set priorities* **-** At any given moment, you have a number of goals all in different states of completion. Deciding which goals are more important, or time-sensitive, than others is crucial. If you have a college interview scheduled on the same day as a training session for your big half-marathon, the interview would take precedent as it is far more time-sensitive, and probably more important.
  - *Keep track of your progress* **-** Journaling is a great way to keep track of both personal and professional progress. Checking in and acknowledging the progress made towards a certain goal is key to staying motivated.
  - *Reward your accomplishments* **-** Acknowledge when you have reached goals and allow yourself to celebrate accordingly.

Writing goals and objectives

- Goals are broad, abstract statements of intent that help create a vision of what you are striving to accomplish.
- Objectives are measurable, specific statements that lead toward program goals and define what change the community will try to achieve.
- The importance of both in anchoring the community health-planning process cannot be emphasized enough, for vague goals and objectives are likely to yield scattered, unfocused efforts.
- Each objective should answer these questions:
    - Who will receive the intervention? (Whose health is its focus?)
    - What health benefit should these persons receive?
    - How much of that benefit should they receive?
    - By when should it be achieved?

For example:

- *By 1998, the prevalence of smoking among county residents aged 18 years and older will be reduced by 15% from 25% (BRFSS 1991 baseline) to 21%.*

- *Our goal is to reduce the number of premature deaths due to heart disease in our community.*

- *By 1998, the rate of fatal injuries among county residents caused by drinking and driving will be reduced by 15% from 7/1,000 to 5/1,000.*

Objectives are active, working tools and not merely academic exercises. An objective:

- Specifies a single key result.
- Specifies a target date.
- Is specific and quantitative.
- Specifies what and when, not why and how.
- Is readily understandable to those involved.
- Is realistic, attainable, yet a challenge.
- Provides limits to expenditures of time and effort.
- Identifies criteria for evaluating achievement.
- Provides orientation to cooperating agencies in the community.

Community health planning uses two types of objectives to clarify community goals:

- Behavioral objectives—for the leading behaviors that contribute to death or disability and are the focus of the intervention
- Intervention objectives—for the interventions you wish to undertake

Writing behavioral and intervention objectives

- Behavioral objectives refer to those behavioral risk factors that contribute to the cause of death specified in your community goals.
- The intervention objectives refer to the intervention activities you plan to undertake.
- The community goal is more general, the behavioral objectives are more specific, and the intervention objectives the most specific. Using the above example, the hierarchy of objectives would include the following:

<u>Community goal:</u>
Our goal is to reduce the number of premature deaths due to heart disease in our community.

<u>Behavioral objectives:</u>

- *By 2000, the prevalence of county residents who smoke will be reduced by 20%, from 32% (1994 BRFSS data) to 25.6%.*
- *By 2000, the prevalence of physically inactive adults in our county will be reduced by 15%, from 38% (1994 BRFSS data) to 32%.*

Intervention objectives:

- *By January 1995, 20% of participants in a "Quit and Win" smoking cessation contest will still be nonsmokers one year after the contest (January 1994).*
- *By January 1996, the rate of onset of smoking among county school students, grades 6-9, will be reduced by 20% from 158/1,000 (1993 YRBS data) to 128/1,000.*
- *By January 1995, the county school board will set policy that allows the community to use school playing fields for evening and weekend recreational activities.*
- *By July 1996, 5 companies that employ 50 or more workers in the community will provide access to programs that address physical activity, good nutrition, and cessation of tobacco use.*

Community health planning process

Although plans can be adapted to various health problems and communities, the phases of the process remain the same. Thus, once the mechanisms of the planning process are in place, only a few modifications are needed to address additional health issues. Phases can be repeated as new health priorities are identified, new target groups are selected, or new interventions are developed. The activities within phases may overlap as the process is carried out. Each of the five phases that constitute plan is described hereafter.

***Step I: Mobilizing the community* -** Mobilizing the community is an ongoing process that starts in step I as a community organizes to begin planning and continues throughout the process. In phase I:

- The community to be addressed is defined,
- Participants are recruited from the community
- Partnerships are formed
- Demographic profile of the community is completed

***Step II: Collecting and organizing data* -** Step II begins when the community members form working groups to obtain and analyze data on mortality, morbidity, community opinion, and behaviors. These data, obtained from various sources, include quantitative data (e.g., vital statistics and survey) and qualitative data (e.g., opinions of community leaders).

***Step III: Choosing health priorities* -** During this phase, objectives related to the health priorities are set. The health priorities to be addressed initially are selected.

***Step IV: Developing a comprehensive intervention plan* -** Using information generated during steps II and III, the community group chooses, designs, and conducts interventions during phase.

***Step V. Build on existing services* -** In order to prevent duplication and to integrate the existing services, the community group identifies and assesses resources, policies, environmental measures, and programs already focused on the risk behavior and to the target group.

***Step VI: Evaluating plans* -** Evaluation is an integral part of the planning process. It is ongoing and serves two purposes: to monitor and assess progress during the five phases of planning and to evaluate interventions. The community sets criteria for determining success and identifies data to be collected.

Community planning materials

This Concept Guide is part of a three-part package of materials designed for the local coordinator, the person who initiates the planning process within a community. This local coordinator will:

- Have major coursework and experience in health education and community health promotion.
- Be able to adapt the planning process to meet the needs of the community.
- Serve as facilitator of this community-based process by working with a broad-based community group and ensuring community ownership.
- Use expertise and resources at the community, state, and federal levels.

Concept guide **-** This Concept Guide presents an overview of plans, followed by separate chapters on each phase of the planning process. The guide includes tools, or forms, for planning and conducting various activities. It also provides background information on topics important to managing plans, such as group dynamics and statistical analysis, as well as practical information and suggestions for tailoring the process to the needs of your community. Develop a one-page tip sheet that relate to the management of plan and group dynamics. You are encouraged to copy and share the tip sheets with group facilitators and working group chairpersons.

Meeting guide **-** The Meeting Guide is intended for use when planning and conducting meetings. It contains an introduction, followed by a separate section for each phase of the planning process. For each phase, it includes meeting objectives, recommends an agenda, and suggests specific activities that can be incorporated with your own ideas.

Visual aids **-** The packet of Visual Aids includes camera, ready copy for overheads and reproducible text for handouts. Some of these materials are intended to serve as models for overheads and handouts you will need to develop to present data and other information for your own community.

Detailed implementation plan

The analysis steps make it possible to draw a Project Planning Matrix (PPM). The PPM is a matrix of four columns and four rows providing 18 squares for a comprehensive description of a project. PPM shows both the project's logical structure (the links between the inputs/activities and the objectives to be achieved under certain Assumptions), and its major quantitative data. The PPM is useful in two ways:

- Showing activities t be carried out in their sequential order
- Shows the type of activity to be done and by who
- Shows what activities should be done in what time
- Shows the budget needed for each activity

Goal: Improve health of the community in Siaya County

Objective	**Activity**	**Time**	**Responsible person**	**Amount**	**Source of cash**	**Threats/ assumptions**
To reduce the number of deaths due to diarrheal diseases from 20 per month to 15 per month by 2015	1.Distribute Chlorine to every homes	Jan	CPHO	30,000	DMOH	
	2.Educate family members in 20 homes in the use of water treatment	March	CHPO	-	-	Rain
	3.Drill 2 boreholes in the community	April	Water engineer	200,000	County treasury	Landscape
	4.Construct pit latrines in 30 homesteads	Aug	Water and sewerage office	90,000	KEWASCO	
	5.Provide ORS to children with diarrhea	Dec	Area nurse	20,000	DMOH	Stock out at KEMSA

Community Health Evaluation

- Community health evaluation is systematic studies conducted to assess how well a program is working in a community and why. Through evaluation, the implementers of community health services can:
    - Create more effective and efficient programs
    - Make better management decisions
    - Support new and innovative approaches and emerging practices
    - Continuously improve existing programs
    - Subsequently improve human health and the environment

- Input indicators – are efforts or what you invest in an activity. These can be resources e.g. money, personnel, time etc.

- Process indicators – are procedures, methods or steps followed during implementation of an activity. A process evaluation assesses whether a program or process is implemented as designed or operating as intended and identifies opportunities for improvement. Process evaluations often begin with an analysis of how a program currently operates. Process evaluations may also assess whether program activities and outputs conform to statutory and regulatory requirements, policies, and program design or customer expectations.

- Output indicators – are immediate outcome or results of an activity

Purposes of evaluation

- Answer questions about projects, policies and programs, particularly about their effectiveness and efficiency.
- Assess how well a program is working and why.
- Implementers can make better management decisions; support new and innovative approaches and emerging practices; continuously improve existing programs; and subsequently, improve human health and the environment.
- Evaluations should help to draw conclusions about five main aspects of the intervention:
    - Relevance
    - Effectiveness
    - Efficiency - Finally, cost-benefit or cost-effectiveness analysis assesses the efficiency of a program. Evaluators outline the benefits and cost of the program for comparison. An efficient program has a lower cost-benefit ratio.
    - Impact
    - Sustainability
- Information gathered provides the basis for the evaluative analysis

Types of program evaluations are there

Design - A design evaluation is conducted early in the planning stages or implementation of a program. It helps to define the scope of a program or project and to identify appropriate goals and objectives. Design evaluations can also be used to pre-test ideas and strategies.

Impact - An impact evaluation is a subset of an outcome evaluation. It assesses the causal links between program activities and outcomes. This is achieved by comparing the observed outcomes with an estimate of what would have happened if the program had not existed (e.g., would the water be swimmable if the program had not been instituted).

Cost-effectiveness - Cost-effectiveness evaluations identify program benefits, outputs or outcomes and compare them with the internal and external costs of the program. Cost-effectiveness evaluations identify program benefits, outputs or outcomes and compare them with the internal and external costs of the program. The impact evaluation determines the causal effects of the program. This involves trying to measure if the program has achieved its intended outcomes, i.e. program outcomes.

Measures of success

There are 4 essential key success measures and one auxiliary measure that are all of great value. The 4 essential key measures are:

- Financial viability - Example: profitability.
- Customer satisfaction - Example: performance on customer satisfaction surveys.
- Employee satisfaction - Example: performance on employee satisfaction surveys.
- Contribution to society - Example: number of trees saved by developing paperless processes.

These measures are important because:

- Financial viability measures organizational survival and growth.
- Customer satisfaction is important because without happy customers the organization will fail.
- Employee satisfaction is important because over the long term it is impossible to have an organization with unhappy employees that has happy customers
- Contribution to society (e.g., environment, ethics, safety, social responsibility) is important because every organization needs more than a simple profit motive to attract and retain the best talent and to sustain itself over time. An organization in which greed is a core value will ultimately devour itself.
- Key operational results are not considered an essential key success measure on the order of the first 4 and therefore are viewed as serving an auxiliary function.

Outcome evaluation

Outcome indicators – are long terms effects as a result of the outputs. Outcome evaluations examine the results of a program (intended or unintended) to determine the reasons why there are differences between the outcomes and the program's stated goals and objectives. Outcome measurement is a systematic way to assess the extent to which a program has achieved its intended outcomes. It helps understand whether the program is effective or not.

Monitoring and Evaluation plan

Objective	Activity	Indicators	Means of verification
To reduce the number of deaths due to diarrheal diseases from 20 per month to 15 per month by 2015	1.Distribute Chlorine to every homes	# of homes supplied with chlorine # of homes with patients with diarrhea # of chlorines distributed per home	Household survey Household survey Distribution list
	2.Educate family members in 20 homes in the use of water treatment	# of homes covered with education on use of water treatment # of family members educated on use of water treatment # of educated family members actually treating water	Education program Education report Home visit
	3.Drill 2 boreholes in the community	# of boreholes drilled in the community # of drilled boreholes has water # of people fetching water from the borehole daily	Survey Survey Survey
	4.Construct pit latrines in 30 homesteads	# of latrines constructed # of homes with latrines # of constructed latrines are in use	Construction report Household survey Household survey
	5.Provide ORS to children with diarrhea	# or ORS sachets provided # of children provided with ORS	S11 form ORS register